

**SUA IntelliLearn: Web Based Learning Management
System with Intelligent Teacher Assistance for Saint
Uriel Academy of Taguig City**

**A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Global City**

**Alayan, Rose Fortune A.
Aniasco, Daniel T.
Exequiel, Jana T.
Paller, Frenz Allen J E.**

May 18, 2026

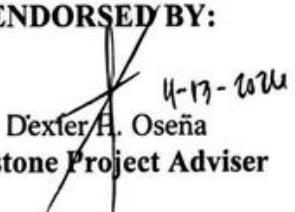
ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: SUA IntelliLearn: An Electronic Learning Management System with Intelligent Teacher Assistance at Saint Uriel Academy of Tagnig City

NAME OF PROPONENTS: Rose Fortune A. Alayan
Daniel T. Aniasco
Jana T. Exequiel
Frenz Allen J E. Paller

In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.

ENDORSED BY:


Dexter H. Oseña
Capstone Project Adviser

APPROVED FOR PROPOSAL DEFENSE:

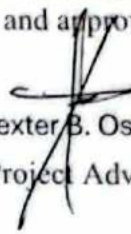

Prince Aaron E. Chavez
Capstone Project Coordinator

NOTED BY:

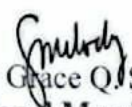

Mark Frederic V. Salonga
Program Head

APPROVAL SHEET

This capstone project proposal titled **SUA IntelliLearn: Web Based Learning Management System with Intelligent Teacher Assistance for St Uriel Academy of Taguig City**, prepared and submitted by **Rose Fortune A. Alayan, Daniel T. Aniasco, Exequiel T. Jana, and Frenz Allen J E. Paller**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.


Dexter B. Oseña
Capstone Project Adviser

Accepted and approved by the Capstone Project
Review Panel in partial fulfillment of the requirements
for the degree of Bachelor of Science in Information
Technology


Melody Grace Q. Sombrero
Panel Member


Edser V. Labiano
Panel Member


Mark Frederick V. Salonga
Lead Panelist

Noted:


Prince Aron E. Chavez
Capstone Project Coordinator


Mark Frederick V. Salonga
Program Head

May 18 2026

TABLE OF CONTENTS

	Page
Title Page	i
Endorsement form for Proposal Defense	ii
Approval Sheet	iii
Table of Contents	iv
Introduction	1
Project Context	1
Purpose and Description	4
Objectives	6
Scope and Limitations	7
Review of Related Literature/Studies/Systems	
Methodology	23
Technical Background	23
Requirements Analysis	28
Requirements Documentation	29
Design of Software, System, Product, and/or Processes	34
References	46
Appendices	
Resource Persons	
Personal Technical Vitae	

INTRODUCTION

Project Context

Many small to medium private institutions still relies on manual or semi-digital processes, which can lead to inefficiencies, data inconsistencies, and limited accessibility of academic information. Although digital platforms existed in various forms earlier, digital education is essentially a modern invention. It is the digitalization of a segment of the educational system. With the development of internet technology, there has been a significant shift in how students communicate and collaborate with academics (Alenezi et al., 2023). The e-Learning Management System (eLMS) has been a major advancement in the development of knowledge management systems. It has been acknowledged and known as an effective and powerful way to support teaching and learning opportunities worldwide as well to deliver education, knowledge, and flexibility to higher education institutions (Muhisn et al., 2022).

Established in June 2005, Saint Uriel Academy of Taguig City offers Junior High School (JHS) and Senior High School (SHS) programs. Saint Uriel Academy is located at Orchid Street, SAMAMA II, Napindan, City of Taguig. The institution operates from 7:00 A.M to 5:00 P.M and currently serves approximately 199 JHS students and 260 SHS students distributed across six sections each, including Accountancy, Business and Management (ABM), General Academic Strand (GAS), Humanities and Social Sciences (HUMSS), Technical-Vocational-Livelihood (TECHVOC), and Information and Communication Technology (ICT) strands. Over the years, the school has experienced steady growth in student population while maintaining a high graduation rate of about 98%. Despite this positive performance, several academic and administrative processes remain manual. Teachers create the class' lesson plan, and syllabus, each with their own manner of teaching, as teachers also manage attendance using SF2 forms. Class programs are encoded in tables by the principal, and most student records are printed, stored in folders, and kept in filing cabinets. While SHS records have had soft copies since 2019, data retrieval and consolidation remain time-consuming.

Communication and academic coordination are primarily conducted through Messenger group chats, where announcements and reminders are posted. However, not all students consistently receive updates, especially those with limited access to social media, resulting in missed submissions or compliance issues. Academic outputs are typically submitted onsite, and learning materials are delivered through printed assignments, books, and PowerPoint presentations during discussions. These fragmented processes create challenges in tracking student progress, managing historical academic records, and ensuring consistent dissemination of information. Research examining interactions between teachers and students on Facebook found that social media communication can blur professional boundaries between academic and personal relationships. Studies have shown that the use of social networking platforms for academic communication between teachers and students potentially raises concern regarding privacy, professionalism, and appropriate interaction in educational environments (Levy et al., 2024).

Integrating learning management systems has emerged as a global trend to increase student engagement and instructional effectiveness. To improve accessibility and enable data-driven decision-making, Philippine educational institutions are increasingly adopting digital platforms. Saint Uriel Academy acknowledges the potential benefits of technology at the institutional level, especially given the growing acceptance of artificial intelligence systems that can help teachers create evaluations and teaching materials. Even while AI can simplify educational activities, it also poses risks, such as a mismatch between learning objectives and generated content, underscoring the need for a well-designed, well-managed system.

Given these conditions, the development of an electronic learning management system tailored to the school's workflow is both timely and necessary. The proponents can undertake this project due to their background in information technology, system design, and software development, enabling them to analyze existing processes and translate them into a functional digital solution. With the ongoing advancement of digital technology in education, schools are increasingly adopting electronic learning management systems to improve instructional delivery, communication, and recordkeeping. It will serve as a centralized platform for announcements, including wide-reaching academic notices,

manage attendance and records, support historical student data tracking, and provide intelligent assistance features for teachers while maintaining administrative control.

The rationale for improving operational efficiency, enhancing communication, and supporting data-driven instruction directly relates to the research problem of how a customized learning management system with intelligent teacher assistance can address the institution's current manual, fragmented processes. This establishes a clear connection between the identified needs of Saint Uriel Academy of Taguig City and the proposed SUA IntelliLearn system, positioning the study as a practical and institution-responsive technological solution.

Purpose and Description

The proposed project, SUA IntelliLearn, is an Electronic Learning Management System (ELMS) developed exclusively for Saint Uriel Academy of Taguig City. The system provides a centralized and intelligent digital platform that integrates academic management, instructional support, and data-driven decision-making into a single, unified environment accessible to administrators, principals, teachers, and students.

SUA IntelliLearn enables teachers to digitally record grades, monitor student academic progress, prepare academic reports, manage attendance, and upload learning materials including documents, videos, and presentations. Students can access academic materials, answer quizzes, view their academic progress, and receive announcements directly through the system. School administrators and principals are provided with tools for managing user accounts, assigning courses, and overseeing institutional academic data in an organized and structured manner. A defining feature of SUA IntelliLearn is its integrated Artificial Intelligence (AI) support tool, designed exclusively for teachers and the school principal. This AI-powered assistant allows teachers to automatically generate quiz questions derived directly from uploaded pdf or text learning materials, ensuring that assessments are aligned with instructional content. Teachers can also interact with the AI chat assistant to receive lesson recommendations, teaching strategies, and subject-related guidance tailored to their specific classroom needs.

The system further incorporates an intelligent student performance monitoring feature that continuously analyzes quiz scores, attendance records, and academic activity data to automatically identify students who may benefit from early academic intervention. Visual performance indicators including charts and graphs are generated to support teachers in making informed instructional decisions. For the school principal, the AI tool automatically summarizes and organizes teacher evaluation results submitted by students and faculty, providing a comprehensive and efficient overview of academic feedback across the institution.

SUA IntelliLearn also offers robust reporting and analytics capabilities, allowing teachers to generate and export academic performance reports in PDF and Excel formats. An integrated notification and announcement system ensures that students, teachers, and administrators receive timely updates regarding quizzes, deadlines, and important school announcements directly within the platform.

What distinguishes SUA IntelliLearn from conventional learning management systems is its integration of intelligent teacher assistance, automated at-risk student detection, and AI-driven evaluation summarization — all customized specifically to the academic structure, curriculum, and operational needs of Saint Uriel Academy of Taguig City. The system represents a purposeful integration of information and communication technology and artificial intelligence into school administration and academic management, designed to be user-friendly, scalable, and sustainable for a growing academic institution.

Objectives

To design and develop SUA IntelliLearn+, a web-based learning management system with AI teacher assistance to improve academic management, monitoring, and instructional efficiency.

Specific Objectives

- To design and develop a centralized academic management module that efficiently handles student records, learning materials, attendance tracking, and learning activities.
- To design and develop an intelligent module that uses AI for quiz generation, student performance monitoring, at-risk detection, and feedback analysis.
- To design and develop a communication and notification module that delivers real-time announcements, alerts, and updates to students, teachers, and administrators.
- To evaluate the developed system using the ISO/IEC 25010 software quality model in terms of functionality, usability, reliability, performance efficiency, and security.

Scope and Limitations

This study focuses on the development of SUA IntelliLearn: An Electronic Learning Management System with Intelligent Teacher Assistance for Saint Uriel Academy of Taguig City. The primary purpose of this study is to design and develop a centralized and intelligent e-learning system that improves the efficiency of teachers, students' engagement, academic monitoring, and decision-making within the institution.

The beneficiaries of this system are the administrators, principals, teachers, and students of Saint Uriel Academy. Administrators, principals, teachers, and students have different privileges in accessing modules in the system. The study was conducted over a period of twelve months, from January 2026 to December 2026 at Saint Uriel Academy of Taguig City. The period involved planning, requirements gathering, system design, system development, testing, and evaluation.

Features:

- **Built-in Attendance**

The Built-in Attendance module has the capability to automatically store the attendance records in a centralized database, which can be easily accessed by the teachers and administrators. The attendance records can also be used by the analytics module to track patterns such as the frequent absenteeism that could be an indication of the student being at risk academically.

- **AI-Powered Quiz Generator**

Teachers can upload lesson materials such as PDF files, documents, or text-based learning content into the system, and the AI tool automatically generates quiz questions based on the uploaded material. The generated questions are designed to align with the instructional content, ensuring that the assessment reflects the topics discussed in class. Teachers may review, edit, or customize the generated questions before publishing the quiz to students.

- **AI-Powered Risk Student Detector**

The system evaluates multiple indicators such as quiz scores, activity submissions, and attendance records. Using these data points, the AI tool detects patterns that suggest declining academic performance. When a student is identified potentially at risk, the system notifies teachers through visual indicators, charts, or alerts.

- **AI-Powered Peer Evaluation Analyzer**

The AI system automatically processes and summarizes the collected responses. It identifies common feedback trends, strengths, and areas that may require improvement. The results are presented in summarized reports, graphs, or categorized insights.

- **Notification System**

The Notification System provides real-time alerts and reminders to users within the platform. Students, teachers, and administrators receive notifications regarding important academic activities such as newly posted quizzes, upcoming deadlines, announcements, or updates from instructors. Notifications appear directly within the system interface, ensuring that users are informed even without relying on external communication platforms such as Messenger.

- **User Profile Display**

The User Profile Display module provides a personalized account page for each system user, including students, teachers, administrators, and the principal. The profile contains basic information such as the user's name, role, assigned classes or subjects, and other relevant academic details. It allows users to easily view their academic responsibilities and personal information within the system.

- **Announcement System**

The Announcements System allows teachers and school administrators to publish official academic announcements directly within the SUA IntelliLearn platform. These announcements may include reminders about quizzes, assignment deadlines, school events, schedule changes, or other important updates. Once posted, announcements become accessible to the intended users, ensuring that all students receive the same information through a centralized platform.

- **AI Chathead Assistant (Uriel AI)**

The AI Chathead Assistant, known as Uriel AI, functions as an intelligent support tool designed specifically for teachers and the school principal. It aids through a conversational chat interface integrated within the system. The assistant can also help explain academic topics, recommend classroom activities, and support teachers in preparing instructional materials.

Limitations

SUA IntelliLearn, a web-based application, is not available as a mobile and desktop application, hence, can be accessed in mobile and desktop platforms via web browsers. The developed AI Teacher Assistance feature of the system should not be used to evaluate subjective outputs of students, such as essays, projects, or performance-based outputs. The AI Quiz Generator feature is limited to creating questions for quizzes based on the syllabus given by teachers. The AI Chatbot does not store conversation history. Every time a user starts a new chat, it is treated as a new conversation session.

Review of Related Literature/Studies/Systems

Foreign

“ICT-Enabled Information Management System (Ims) in Educational Organizations (Mishra & Purohit, 2025) “

According to Mishra & Purohit (2025), the fast development of Information and Communication Technology (ICT) has significantly influenced educational institutions to adopt ICT-enabled Information Management Systems (IMS) to improve administrative and academic processes. These systems enhance data handling, streamline communication, and enable data-driven decision-making, allowing schools to transition from traditional management approaches to more efficient digital structures. ICT-based systems support performance monitoring, digital resource management, and real-time institutional insights, which contribute to improved learning outcomes and more effective school operations. Similar to ICT-enabled IMS, the SUA IntelliLearn aims to digitalize and automate the management of academic records, learning materials, and class data, which are currently handled manually through paperwork and traditional record-keeping methods. By implementing an LMS, the system can streamline the storage, organization, and retrieval of academic information, thereby reducing the risks of data loss, record inaccuracies, and time-consuming administrative processes.

“Academic Records Management Practices in Africa Higher Education Institutions: Empirical (Yidana, P., & Akuna E., 2025)”

Review

According to Yidana & Akuna (2025), all academic information related to individuals, maintained in various formats and media by education institutions, is highlighted in students' academic records. By reviewing how academic documentation is currently managed across the African tertiary sector, this study highlights that the integration of digital filing solutions drastically optimizes the speed and reliability of record-keeping processes. Additionally, the review emphasizes that African education authorities are Taking proactive measures to protect students' academic records. SUA IntelliLearn aims

to facilitate the management of students' academic records, pertinent to the study conducted by Yidana & Akuna (2025). With the implementation of a system that has the capability to manage student academic records and learning materials, it maintains high data integrity, minimizing human error, the loss of academic documents, and ensuring the accuracy of student academic records.

“Perceptions of Organizational Change Readiness for Sustainable Digital Transformation: Insights from Learning Management System Projects in Higher Education Institutions

(Veseli, A., Hasanaj, P., & Bajraktari, A. 2025)”

The adoption of Learning Management Systems (LMS) in educational institutions has been shown to enhance operational efficiency, improve educational quality, and support the digital transformation of academic processes. A study conducted across multiple higher education institutions highlighted that LMS implementation not only automates administrative and academic functions but also promotes more effective management of student records, class data, and performance tracking. The research emphasized that institutional readiness, leadership support, and alignment with organizational goals are critical factors in the successful adoption of such systems, particularly in contexts where traditional, manual methods are still prevalent. These findings support the development of the proposed Electronic Learning Management System (ELMS), which aims to automate and streamline the management of student records, learning materials, and academic performance, replacing inefficient paper-based systems and enabling teachers and administrators to manage school operations more effectively. By incorporating lessons from LMS adoption in higher education, the proposed ELMS addresses both the technical need for automation and the organizational requirements necessary for smooth implementation within the school environment.

“Native Apps vs. Progressive Web Apps: A Comparative Analysis of User Experience and Development Costs

(Samsyudin, I. 2025)”

According to Samsyudin, (2025), recent advancements in application development have introduced two dominant approaches: native applications and Progressive Web Apps (PWAs). A study comparing these technologies evaluated key factors such as performance, cost-efficiency, user experience, load time, offline functionality, user engagement, and overall user satisfaction. The findings revealed that native applications generally perform better in terms of speed, responsiveness, and offline capabilities, making them suitable for high-performance industries such as gaming and banking. The study highlighted that web-based applications, particularly PWAs, offer greater overall efficiency and practicality compared to native applications. One of the primary advantages of PWAs is their cost-efficiency, as they require only a single codebase that can operate across multiple platforms, including both mobile and desktop devices. This diminishes the need to develop separate applications for different operating systems, significantly lessening development time, maintenance efforts, and overall costs. The study also acknowledges certain limitations of web-based applications. Compared to native apps, PWAs may have slower load times, limited offline functionality, and reduced access to device-specific features, which can affect performance and user engagement in more demanding applications. These drawbacks make native applications fitter for systems that require intensive user interaction and high performance. Adopting a web-based application supports the objective of developing a system that is accessible on both mobile and desktop platforms while minimizing development and maintenance costs. Although there are some trade-offs in performance, the benefits of cost-efficiency, accessibility, and cross-platform functionality make web-based applications a more practical and sustainable choice for many modern systems.

Local

“Effectiveness of Enhanced Basic Education Information System (EBEIS)/ Learner Information System (LIS) Data and School Management

(Lopez, A., & Rufino Lorejo, 2023)”

Lopez & Rufino Lorejo (2023) stated that in providing available data and information, creating an online information system should be prioritized for allocating resources and operational targets. The study’s findings revealed that coordinator, classroom teachers, and administrators were satisfied with using data in school management, the information literacy provided, and the ICT facilities used for online access. The study also indicated that some schools require much better internet connectivity and more streamlined manuals for system use. Directly relevant to SUA IntelliLearn, the study validates the transition from manual to ICT-based management focusing on data management of student academic records.

Foreign

“Comprehensive Analysis of ICT-Based Learning Management Systems: Best Practices for Enhancing Digital Learning Experiences

(Vijoriya et al., 2026)”

According to Vijoriya et al. (2026), Learning Management Systems (LMS) have become essential ICT tools that enhance the teaching and learning process in modern educational institutions. Studies have shown that LMS platforms improve course delivery, user engagement, and access to learning materials through digital platforms. The study’s main discovery showed that the enhancement of Artificial Intelligence (AI) features such as personal learning recommendations, automate quizzes, test and quizzes, and other forms of tests. This yields massive improvement, user engagement, and satisfaction. The study supports the proposed system, SUA IntelliLearn, which is the integration of Artificial Intelligence (AI) features within LMS, such as automated quiz generation, personalized learning recommendations, and data analytics. The study discusses the implementation of

ICT-based Learning Management Systems (LMS) and highlights how these systems improve the teaching and learning process through digital learning platforms.

“Automated AI-driven System for Early Detection of At-risk Students

(Embarak, O. H., & Hawarna, S., 2024)”

According to Embarak and Hawarna (2024), machine learning algorithms are used to analyze data combinations and identify patterns that predict a student’s likelihood of falling behind. Building on this concept, the study shows the development of an automated system that is made to know at-risk students that incorporates an artificial intelligence–based response using explainable artificial intelligence (XAI) to identify those who may fall behind or drop out of school. The study found that the system effectively uses identified features to continuously predict students’ performance, enabling advisors, administrators, and students to have proactive measures in maintaining high academic progress. In a similar way, SUA IntelliLearn provides teachers with analytics that help identify students who are at risk of failing a specific course.

“AI Chatbot for Teachers for an LMS Platform

(Cardona, C. R., et al., 2024)”

According to Cardona et al. (2024), a chatbot agent was developed using Python, RASA, and PHP to automate teacher assistance within the learning management platform Ciudad Educativa, which supports primary and secondary education. The chatbot was integrated into the platform’s communication module and evaluated by teachers, with results showing that 82.9% of the chatbot’s responses were considered applicable. Similar to the system proposed by Cardona et al. (2024), the chatbot in SUA IntelliLearn can help teachers quickly ask for help with creating quizzes and activities, and with how they would implement lessons in class, all based on the syllabus they provide. This reduces teachers' workload, improves the efficiency of managing class-related tasks, and allows them to focus more on instruction rather than on administrative responsibilities.

Foreign

“Students’ Perception of the Use of Canvas-Based Learning Activities in Online Classroom: Focusing on Its Benefits

(Asri Siti Fatimah, Bilqis Nurfitriani, & Ruslan, 2022)”

The integration of Learning Management Systems (LMS) in education has significantly improved students’ access to academic resources, communication, and participation in learning activities. According to Alenezi et al. (2022), advances in internet technologies have transformed how students interact with academic content and instructors, enabling more flexible and collaborative learning environments. LMS platforms provide students with centralized access to learning materials, announcements, and assessments, which enhance their overall learning experience. Digital assignment and assessment submissions streamline efficiency and convenience. Online systems let students avoid paperwork and submit work anytime, anywhere with internet access. This flexibility supports modern educational environments focused on digital learning. The development of SUA IntelliLearn incorporates LMS functionalities similar to those tailored to Saint Uriel Academy's needs. By enabling students to upload assignments, access academic materials, receive announcements, and complete quizzes within a centralized platform, the system directly addresses the challenges of fragmented communication and manual processes identified in the institution. Consistent with existing literature, these features are expected to improve engagement of students, improve the ability to access information, and support self-directed learning.

“The Effectiveness of Learning Management System (LMS) as a Digital Tool for TESL Undergraduates at Higher Education Institution in Malaysia

(Mariappan K., Singh Kaur A.G., & Singh Kaur P.B., 2024)”

According to Mariappan, et al. (2024), Learning Management Systems (LMS) is known and recognized as effective digital tools in helping and supporting student learning and participation in online environments. A study conducted among Teaching English as a Second Language (TESL) undergraduates examined the effectiveness of LMS as platform for delivering course content and facilitating academic activities. One of the most used

LMS platforms, Moodle, was highlighted for its suitability in online learning due to its diverse and interactive features. These features include access to instructional materials, video resources, assignment submission, external links, and online quizzes, all of which contribute to a more engaging learning experience for students. According to the research findings, integrating learning activities via a management system significantly bolstered student engagement levels and improved the efficiency of academic assessments. Students reported that the LMS environment made it easier for them to engage with course content, complete assignments, and follow lessons in a structured and organized manner. The availability of online activities such as quizzes and project submissions allowed students to actively participate in the learning process, even outside the traditional classroom setting. These findings support the inclusion of student-centered modules in the SUA IntelliLearn system, particularly those that allow students to upload assignments, access learning materials, view announcements, and complete quizzes online. Like the LMS discussed in the literature, these features are expected to enhance student participation, improve accessibility to academic content, and provide an efficient and more engaging learning environment within Saint Uriel Academy.

Local

***“Enhancing E-Learning System Through Learning Management System (LMS) Technologies: Reshape the Learner Experience
(Abaricia, C., & Delos Santos, M. L., 2023)”***

Research conducted by Abaricia & Delos Santos (2023) leveraged data mining techniques to analyze how an integrated digital learning gateway influences student engagement and satisfaction. The study showed that personalized learning trackers help students monitor their progress. This feature increases self-awareness and motivates active learning. Collaboration tools, such as communication with instructors and peers, were seen as encouraging an interactive, supportive environment. Further, user-friendly tools and clear progress display help increase involvement and motivation. These findings support the development of student-oriented modules in the SUA IntelliLearn system. Features such as access to academic materials, assignment submission, quiz participation, and announcement viewing align with the functionalities identified in the literature. By

incorporating these elements, the system is expected to improve student engagement, provide flexible learning opportunities, and enhance academic monitoring, thereby addressing the current limitations of Saint Uriel Academy's manual fragmented processes.

Review of Related Systems:

Chamilo LMS for Web-Based E-Learning Development in a Vocational High School



Figure 1: Chamilo LMS UI

Chamilo LMS is a web-based, open-source learning management system for the creation of a structured e-learning environment for educational institutions. This research was conducted at SMK Negeri 4 Padang. The study was on the application of Chamilo LMS to improve online learning and academic management in the school. Teachers can upload learning content, run quizzes, manage assignments, run discussion forums, and track student progress through a centralized digital system. It also provides mobile accessibility which enables students to access educational resources anytime, anywhere. The researchers stressed that the use of Chamilo LMS improved the delivery of instruction, facilitated better communication between teachers and students, and provided a more systematic manner of managing online learning activities.

i-LNHS: A Learning Management System for Lananpin National High School



Figure 2: i-LNHS LMS UI

i-LNHS is a Moodle-based learning management system developed for Lanangpin National High School to provide distance learning and improve the monitoring of students' academic performance. The system was developed during the COVID-19 pandemic when schools shifted from traditional classroom learning to online and modular learning. i-LNHS provides teachers with a centralized e-learning platform to share learning materials, obtain student assignments, conduct assessments online, and track academic progress. The system also provides students access to self-learning modules, course materials and academic resources anytime through the internet. According to the researchers, the implementation of i-LNHS improved accessibility, enhanced communication, and addressed the shortcomings in the traditional modular learning process.

TelEducation: Customized Learning Management System for Isabela State University–Ilagan Campus

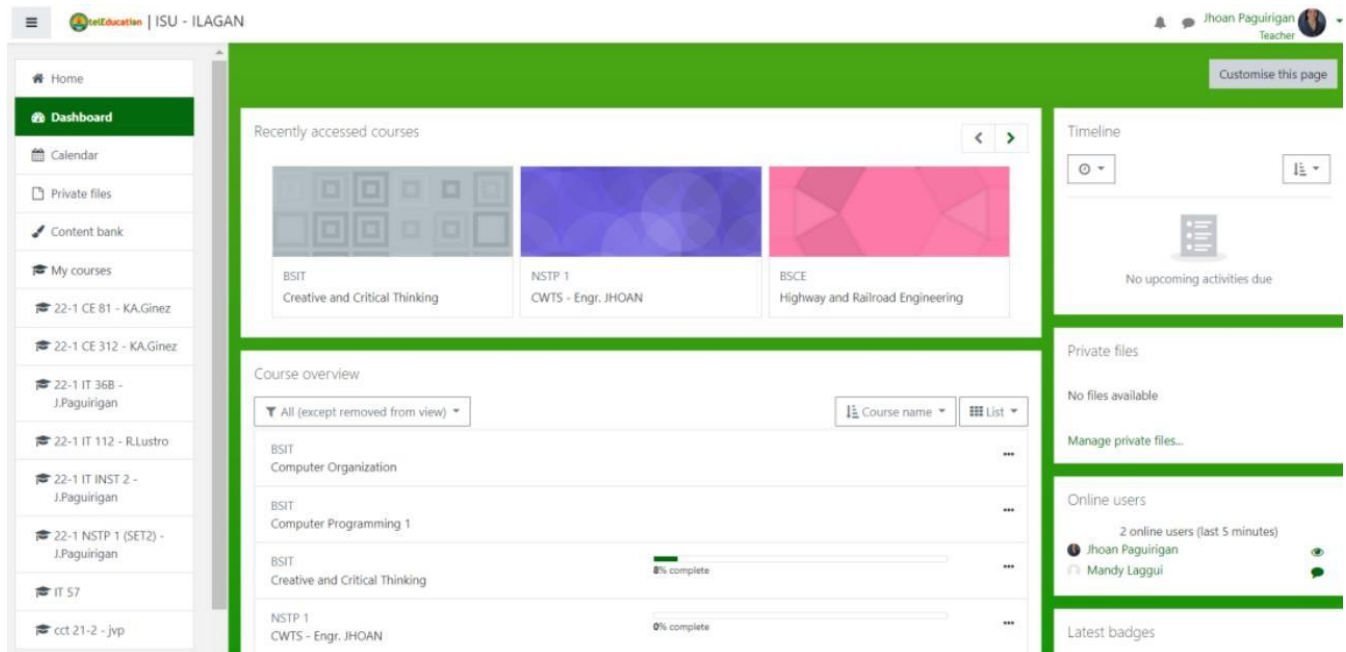


Figure 3: TelEducation LMS UI

TelEducation is a Learning Management System developed for Isabela State University–Ilagan Campus, that is designed to accommodate flexible teaching and learning modalities amidst the COVID-19 pandemic. The platform was developed to offer faculty, staff and students a centralized digital learning environment, leveraging the institution’s existing ICT infrastructure. Teachers can use TelEducation to upload learning content, prepare quizzes and assignments, conduct online discussions and monitor students’ participation and academic performance. The system also provides for collaborative learning and online communication between users with integrated learning tools and resources. The study found that the customized LMS enhanced accessibility, facilitated flexible learning, and enhanced the institution’s ability to provide education in an online environment.

Synthesis

The examined literature and studies establish that ICT-enabled Information Management Systems (IMS) are vital for modern education, highlighting that the shift to digital systems, as shown by Mishra and Purohit (2025) and Veseli et al. (2025), drives efficiency and better decision-making. Digitizing academic records, as discussed by Yidana and Akuna (2025) and Lopez and Rufino Lorejo (2023), is a key approach for protecting data and preventing errors. SUA IntelliLearn utilizes these principles at Saint Uriel Academy to secure data.

Recent research further shows that Learning Management Systems are evolving from basic content storage to intelligent, dynamic platforms with AI functions—automated quizzes, chatbots, and predictive analytics—demonstrated by studies from Asri Siti Fatimah, Mariappan, Vijoriya et al., Cardona et al., Embarak and Hawarna, and Abaricia and Delos Santos. SUA IntelliLearn integrates these advances into its platform by automating record management and providing advanced AI tools to aid teachers and administrators.

Choosing a web-based platform is essential for sustainable system performance and supports accessible transformation, as noted by Samsyudin (2025). SUA IntelliLearn adopts this approach to meet requirements for minimal maintenance, device accessibility, and technological advancement. Chamilo LMS, i-LNHS, and TelEducation, reviewed in this paper, demonstrate how web-based, centralized platforms are crucial for modernizing instructional delivery and academic administration.

Their adoption, spurred by growing demand for online learning post-pandemic, underscores the importance of digital platforms in meeting evolving educational needs. Chamilo LMS at SMK Negeri 4 Padang allows teachers to upload resources, manage assignments, facilitate forum discussions, and track student progress, providing accessible learning beyond traditional classrooms.

i-LNHS enables teachers at Lananpin National High School to organize distance learning, share resources, collect assignments, assess performance, and enhance achievement monitoring, with the main benefit of centralizing all digital education tasks and improving accessibility. The systems share strengths in resource centralization, accessibility, and teacher-student interaction, but lack advanced analytics and automated intelligence. These limitations accentuate the need for SUA IntelliLearn, which enhances current benefits and introduces intelligent, monitoring-driven features to address existing gaps

Feature	<i>Chamilo LMS</i>	<i>i-LNHS</i>	<i>TelEducation</i>	<i>SUA: IntelliLearn</i>
Learning Material Distribution	✓	✓	✓	✓
Quiz and Assessment Management	✓	✓	✓	✓
Grade Recording	✓	✓	✓	✓
Announcements and Notifications	✓	✓	✓	✓
Role-based User Access	✓	✓	✓	✓
Student Performance Reports	✓	✓	✓	✓
Reporting and Analytics	✓	✓	✓	✓
Attendance Monitoring	✓	✓	✓	✓
AI Quiz Generator	X	X	X	✓
AI Teacher Chat Assistant	X	X	X	✓
Student At-Risk Detection	X	X	X	✓
Teacher Evaluation Summarizer	X	X	X	✓
Available to Any Institution	✓	✓	✓	X
Dedicated Mobile Application	X	✓	✓	X
Offline Access	X	✓	✓	X

Table 1: Similarities and Differences

METHODOLOGY

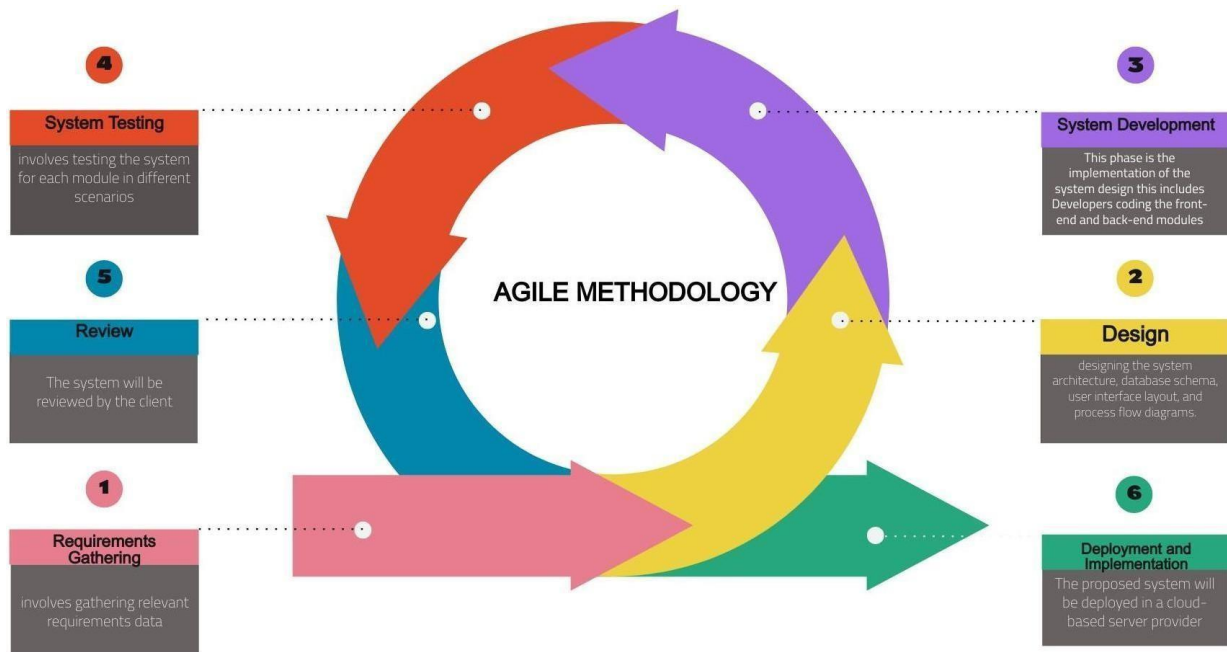


Figure 4: Methodology Model

The developers chose the Agile methodology for the development of SUA IntelliLearn: An Electronic Learning Management System with Intelligent Teacher Assistance, since it allows the system modules to be developed in small and manageable phases while continuously improving the system based on feedback from the students, teachers, and principal of Saint Uriel Academy of Taguig City.

Agile methodology follows a flexible, iterative development process, developing the system gradually through repeated cycles. In this approach, developers first gather the relevant requirements needed to complete the system. Each feature of SUA IntelliLearn is then divided into smaller processes or modules to make development more organized and easier to manage.

The developers will first design the initial user experience (UX), user interface (UI), and system architecture of a module. After completing the design phase, the next step is the development phase, where the developers implement the design by writing the necessary code to build the module.

Once the module has been developed, it will undergo system testing to ensure that it functions properly and meets the specified requirements. If the module satisfies the required functionality, it will be presented to the client for review and feedback. If the client is not satisfied with the output, the developers will return to the earlier stages of design and development to make the necessary improvements.

This process will continue in repeated development cycles, known as sprints, until the client is satisfied and the system meets the intended requirements of Saint Uriel Academy of Taguig City

Technical Background

- **Technologies to be Used**

The proposed system, SUA IntelliLearn, will be developed using modern web-based technologies and techniques to guarantee efficiency, accessibility, scalability, and user experience. The proposed web application will implement a client – server-based architecture where users can access the system services through web browsers such as Google, Firefox, Microsoft edge etc.

The front end will be developed using Web languages such as **HTML, CSS, and JavaScript** to provide modern, user-friendly, and responsive web pages for the clients. The back end will be developed using **PHP** that will handle system logic, data processing, and user authentication. The proposed System will utilize the Relational Database Management System, which is **MySQL Database**. This will be used to perform CRUD operations such as Creating, Reading, Updating, and Deleting Users accounts. To support AI features of the system, the developers will utilize **OpenRouter** a unified API marketplace that provides developers with access to a wide variety of LLMs through a single, standardized API key. This will allow users to access a free open source ai model that can generate, answer questions, and analyze data from the system.

The system will be hosted on a local host server using **Xampp** during the development and testing phase of the project and will be deployed in cloud once it's ready

- **Calendar of Activities**

TASK	START	END	January				February								March								May				June	July	August	September	October	November	December		
			1/27	1/28	1/29	1/30	2/3	2/4	2/5	2/7	2/8	2/9	2/10	2/11	2/18	3/3	3/13	3/24	3/25	3/26	4/7	4/8	4/9	4/10	5/10	5/12	5/19	5/21							
Title page	27-Jan-26	3-Feb-26																																	
Purpose and Description	28-Jan-26	5-Feb-26																																	
Objectives	3-Feb-26	26-Mar-26																																	
Scope and Delimitaion	4-Feb-26	26-Mar-26																																	
Review of Related Literature/Studies/System	5-Feb-26	4-Apr-26																																	
Methodology	6-Feb-26	9-Apr-26																																	
Technical Background	7-Feb-26	26-Mar-26																																	
Requirements Analysis	8-Feb-26	11-Feb-26																																	
Requirement Documentation	9-Feb-26	13-Mar-26																																	
Design of Software	25-Mar-26	10-Apr-26																																	
Reference	10-May-26	12-May-26																																	
Appendices	19-May-26	21-May-26																																	

On progress

Completed

System analysis and Requirements gathering – This activity involves gathering relevant requirements data by interviewing and consulting with teachers, students, and administrator of St Uriel Academy of Taguig City. The goal of this phase is to identify existing problems, functional requirements, and non-functional requirements for the proposed system.

System Design – This involves developers designing the system architecture, database schema, user interface layout, and process flow diagrams.

System Development – This phase is the implementation of the system design this includes Developers coding the front-end and back-end modules, AI integration, and database schema implementation.

System Testing – This involves testing the system for each module in different scenarios. The developers must attempt to get the system to fail and to violate process and security protocols that will enable developers to identify and fix problems that are present in the system. The developers will also perform a user acceptance testing where real users or clients validate that a system meets business requirements and works in real-world scenarios before going live.

Deployment and Implementation – After an accomplished testing, the proposed system will be deployed in a cloud-based server provider and will be introduced to teachers and student in St Uriel Academy of Taguig City. Essentials modification will be based on client's feedback.

Maintenance and Documentation – Minor system updates and maintenance will be conducted as needed to ensure continuous system performance. User manuals and System/Documentation Specific will be prepared to help users to navigate and understand the systems functionalities and help future developers.

- **Resources**

Devices	Purpose	Minimum Specifications
Desktop/Laptop	Used by developers for system development, coding, testing, and documentation. It is also used by teachers, students, and administrators to access the SUA IntelliLearn system through a web browser.	Intel Core I3 or equivalent 4GB RAM 500MB Free disk space
Mobile Device	Used by developers to test the mobile interface and user experience of the mobile version of the web application. It also allows users to access the system through smartphones or tablets.	Android 8.0 higher / IOs 13 or higher 4 GB RAM 200 MB free storage
Printer	Used to print reports, system documentation, students' records, and other academic-related information generated by the system.	Inkjet 600x600 dpi 10 pages per minute USB or Wi-Fi

Table 2: Hardware Resources

Software	Version	Purpose
Windows Operating System	Windows 10	Serves as the main operating system used for development, testing, and deployment of the web-based system.
XAMPP	Version 8.2.x	Provides a local server environment that includes Apache, PHP, and MariaDB/MySQL for hosting and testing the web application locally.
MySQL / MariaDB	MySQL 8.0+ / MariaDB 10.11+	Database management system used to store system data such as user accounts, course materials, and system records.
Visual Studio Code	Version 1.90+	Source code editor used to develop and manage the system's front-end and back-end code.
PyCharm	Version 2024.3+	Used by developers in developing intelligent assistant and AI-related scripts.
Google Chrome	Version 125+	Web browsers are used for testing and accessing the web application interface.
Git	Version 2.45+	Version control system used to track code changes and manage project collaboration.

Table 2.1: Software Resources

Requirements Analysis

The manual way of operating day-to-day tasks affects the entities within the academic facility. Students, teachers, and school administrators struggle to gain more efficient ways of managing activities. The SUA Intellilearn: An Electronic Learning Management System with Intelligent Teacher Assistance is designed to support the school's transition from manual academic processing to a computerized system, thus aiming to achieve a much more streamlined adaptation in terms adapting to the new normal method of education.

The system will mainly be used by the teachers, students, and school administrators to manage the academic process that is happening in the daily operation of Saint Uriel Academy of Taguig City. Quizzes, grading, and recording of the grades will be managed by the teachers. School administrators as well as the IT department will be the ones responsible for managing the registering of new users of this system.

The intent of the developed framework is to modernize and automate the manual complexities of distributing the quizzes, the grading of activities, grade reporting, and attendance monitoring. This will serve as the replacement of doing the arduous processes that all the involved users experience in the current method of operation.

The system will operate in an online environment and can be accessed using computers, laptops, or mobile devices with internet connectivity, allowing users to interact with the system anytime and anywhere within the school's learning environment. This system will always be available to allow students and teachers to do tasks during their most available time of the day on either weekends or academic days.

The current manual procedure involves teachers preparing quizzes on paper, distributing them to students, collecting answer sheets, manually checking or encoding scores, and storing results in Excel files. The proposed ELMS will digitize this workflow by allowing teachers to create quizzes in the system, students to answer online, and the system to automatically record and store data securely in a centralized database.

Requirements Documentation

Functional Requirements

1. User Management

- **REQ001:** The system shall allow user registration and login for administrators, teachers, and students.
- **REQ002:** The system shall support role-defined architecture where access to tools and records is strictly governed by whether a user is classified as an admin, teacher, or a student.
- **REQ003:** The administrative interface shall grant administrators to create, modify, and deactivate user accounts.

2. Learning Management Features

- **REQ004:** The system shall support the distribution of educational content by allowing teachers to upload learning materials (PDFs, videos, presentations), activities, and quizzes.
- **REQ005:** The system shall have a digital ledger to document and track the daily presence of students within the classroom
- **REQ006:** The system shall allow students to view, answer, submit, and download course materials.
- **REQ007:** The system shall allow Principal to create, edit, and delete courses or subjects and assign teachers to a course.
- **REQ008:** The administrative module must facilitate the registration of students in subjects assigned by the system.

3. Quiz and Assessment Management

- **REQ009:** The system shall allow teachers to create quizzes manually.

- **REQ010:** The system shall allow teachers to use AI-powered quiz generator that automatically generates quiz questions based on uploaded learning materials.
- **REQ011:** The system shall allow teachers to set quiz parameters (difficulty level, number of questions, time limit).
- **REQ012:** The system shall automatically grade objective-type quizzes.
- **REQ013:** The system shall display quiz results and scores to students and teacher

4. AI Chat head (Intelligent Assistant)

- **REQ014:** The system shall provide an artificial intelligence conversational assistant that must be integrated into the primary dashboard that can be seen in the bottom-left region of the display.
- **REQ015:** The system shall allow teachers to query the system for teaching assistance, such as lesson suggestions and quiz explanations.
- **REQ016:** The system shall restrict the AI assistant to school-related and academic queries only.
- **REQ017:** The system shall provide context-aware responses based on uploaded learning materials

5. Student at Risk Detection

- **REQ018:** The system shall monitor student performance data such as quiz scores, attendance, and activity logs.
- **REQ019:** The system shall analyze data to identify students at risk of failing.
- **REQ020:** The system shall generate risk alerts and reports for teachers.
- **REQ021:** The system shall display student performance trends using visual indicators (charts/graphs).

6. Reporting and Analytics

- **REQ022:** The system shall generate academic performance reports for students.
- **REQ023:** The system shall generate class performance summaries for teachers.
- **REQ024:** The system shall allow teachers to export reports in PDF or Excel format.
- **REQ025:** The system shall allow administrators to view overall system usage statistics.

7. Notifications and Announcements

- **REQ026:** The system shall allow teachers and administrators to post announcements.
- **REQ027:** The system shall notify students about new quizzes, deadlines, and announcements.
- **REQ028:** The system shall send alerts to teachers for students identified at risk

8. Data Management

- **REQ029:** The system shall store sensitive information, student records, grades, and activity logs securely.
- **REQ030:** The system shall allow the admin to perform backup and recovery mechanisms for data protection.

Non-Functional Requirements

1. Performance

- **REQ031:** The system shall respond to user requests within 2–3 seconds under normal load.
- **REQ032:** The AI chat head shall generate responses within 8 seconds.
- **REQ033:** The system shall support simultaneous access by multiple users without performance degradation.

2. Security

- **REQ034:** The system shall require validation of user identities and restrict| access based on assigned roles.
- **REQ035:** The system shall encrypt sensitive user data.
- **REQ036:** The system shall require implementing protective barriers that isolate.
- **REQ037:** The system shall guarantee that no user activity goes unlogged, ensuring complete audit trail integrity at all times.

3. Usability

- **REQ038:** The system shall have a front-end that is engineered for ease of use and intuitive interface.
- **REQ039:** The system shall be accessible on desktop and mobile devices.
- **REQ040:** The AI chat head shall be easy to locate and use.
- **REQ041:** The system shall provide clear error messages and guidance.

4. Reliability

- **REQ042:** The system shall remain fully functional and accessible throughout the academic day, supporting peak traffic without interruption.
- **REQ043:** The system shall uphold the precision of its records, preventing any conflicting entries or data corruption during processing.
- **REQ044:** The system shall recover gracefully from system failures or crashes.

5. Scalability

- **REQ045:** The system shall support an increasing number of users and subjects.

6. Maintainability

- **REQ046:** The system shall follow modular system architecture.
- **REQ047:** The system shall be designed for streamlined maintenance, supporting the rapid deployment of software patches and functional requirements.
- **REQ048:** The system shall provide system documentation for developers and administrators.

7. Compatibility

- **REQ049:** The system must ensure full functional parity and visual consistency across all modern web browsers.
- **REQ050:** The system shall support common file formats (PDF, DOCX, MP4).

8. Ethical and AI Limitations

- **REQ051:** The AI assistant shall avoid generating harmful or inappropriate content.
- **REQ052:** The system shall ensure fair and unbiased analysis in detecting at-risk students.
- **REQ053:** The AI system shall maintain data privacy compliance with educational standards.

Design of Software, System, Product, and/or Processes

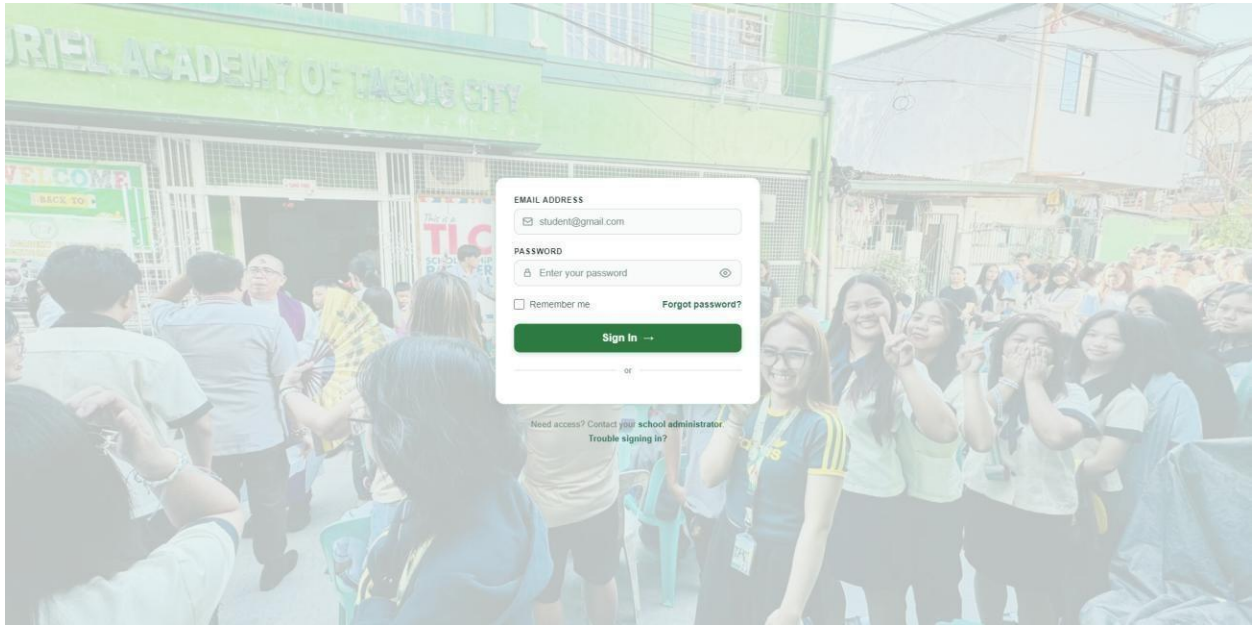


Figure 5: Login page

The login page serves as the entry point for all users of SUA IntelliLearn. It features an email address and password field, a "Remember Me" checkbox, and a "Forgot Password" link for account recovery. At the bottom, a prompt directs new users to contact the school administrator for account creation, reinforcing that self-registration is not allowed only authorized users to gain access. This ensures system security and role-based control from the very start.

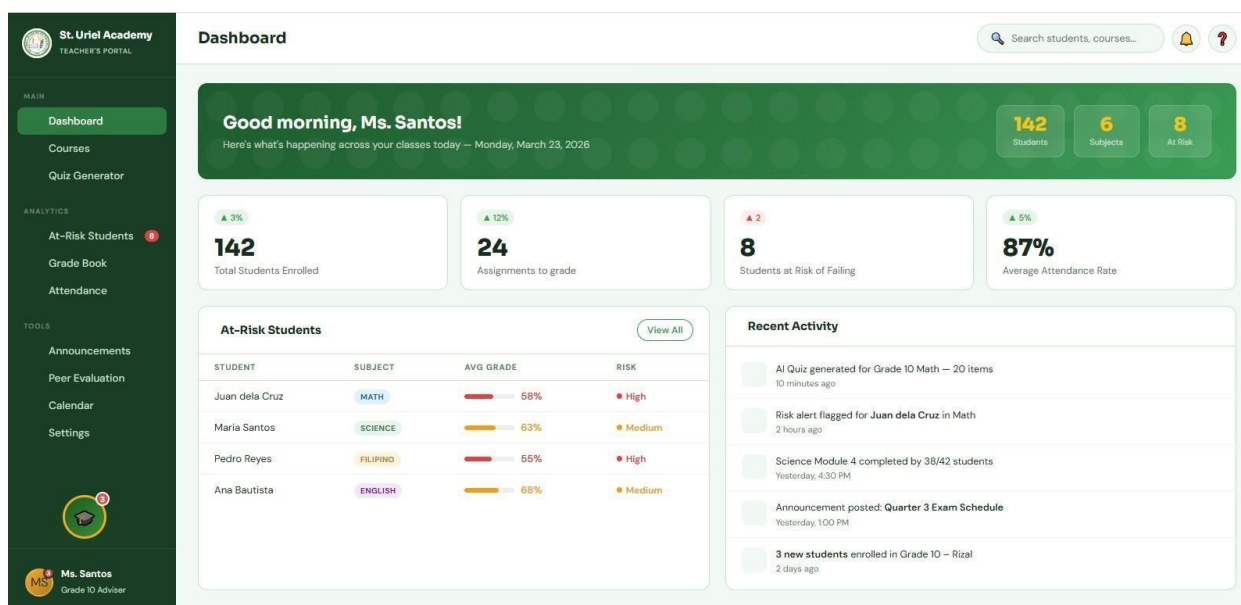


Figure 5.1: Teachers Dashboard

The Teacher Dashboard is the central hub for all teacher activities. Upon logging in, the teacher is greeted with a personalized welcome and a real-time summary of their classes. The dashboard displays four key metrics immediately: total students enrolled, assignments pending for grading, students currently flagged as at-risk, and the average class attendance rate. Below are two main panels. The At-Risk Students panel lists students by name, subject, average grade, and risk level (High/Medium), allowing teachers to immediately identify who needs intervention. The Recent Activity panel on the right logs of system events such as AI-generated quizzes, risk alerts, student completions, and new announcements — keeping the teacher informed without having to navigate away from the dashboard. The left sidebar gives quick access to Courses, Quiz Generator, At-Risk Students, Grade Book, Attendance, Announcements, Peer Evaluation, Calendar, and Settings. The AI chat head (Uriel AI) is also visible at the bottom-left corner.

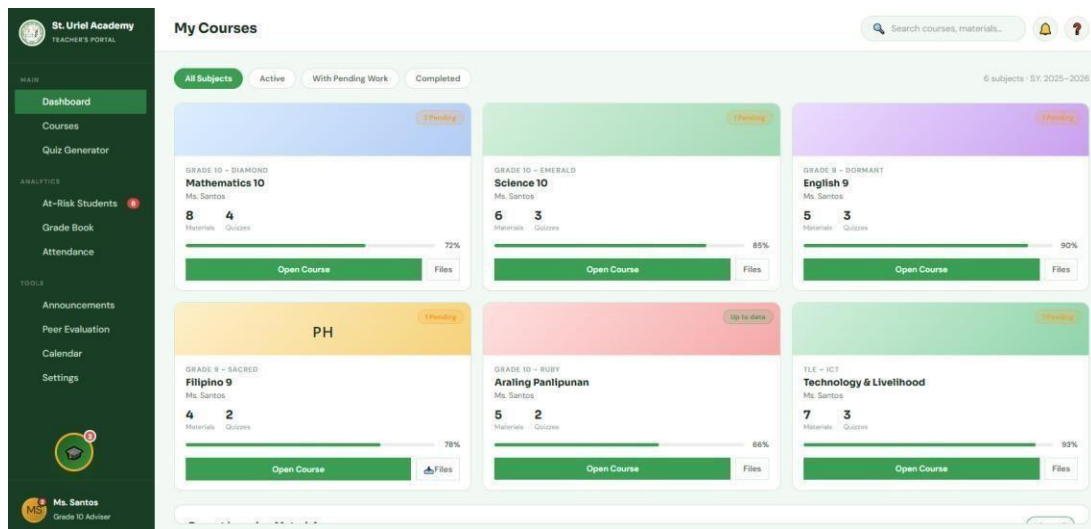


Figure 5.1.1: Teacher's Courses

The Teacher's Courses module allows the teachers to view the course progress of the specific section of learners as well as opening the files and uploading necessary documents for the course.

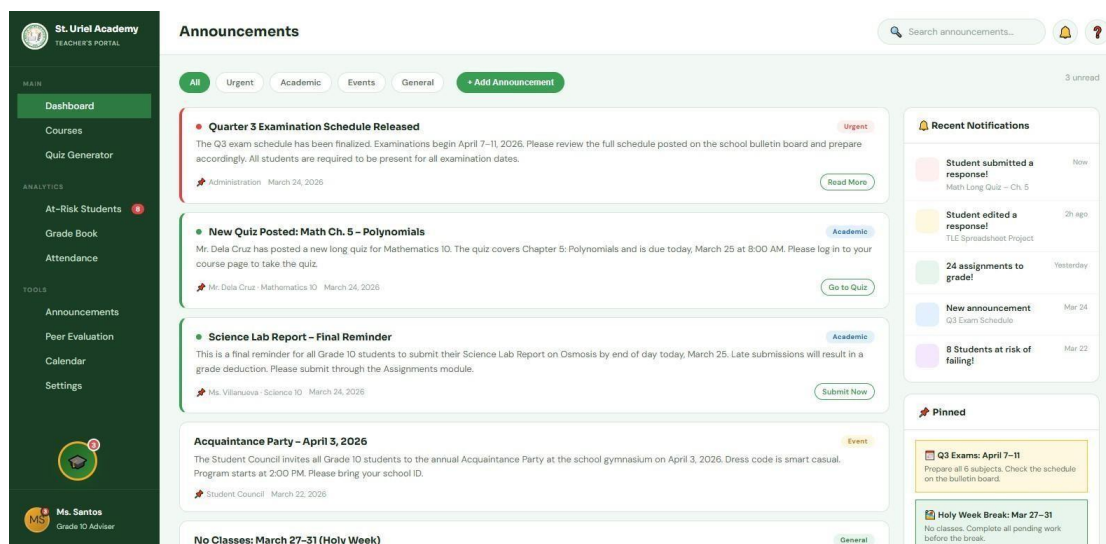


Figure 5.1.2 Teacher's Announcements

The Teacher's Announcement module allows the teachers to view and publish announcements related to deadlines, quizzes, and other course related announcements.

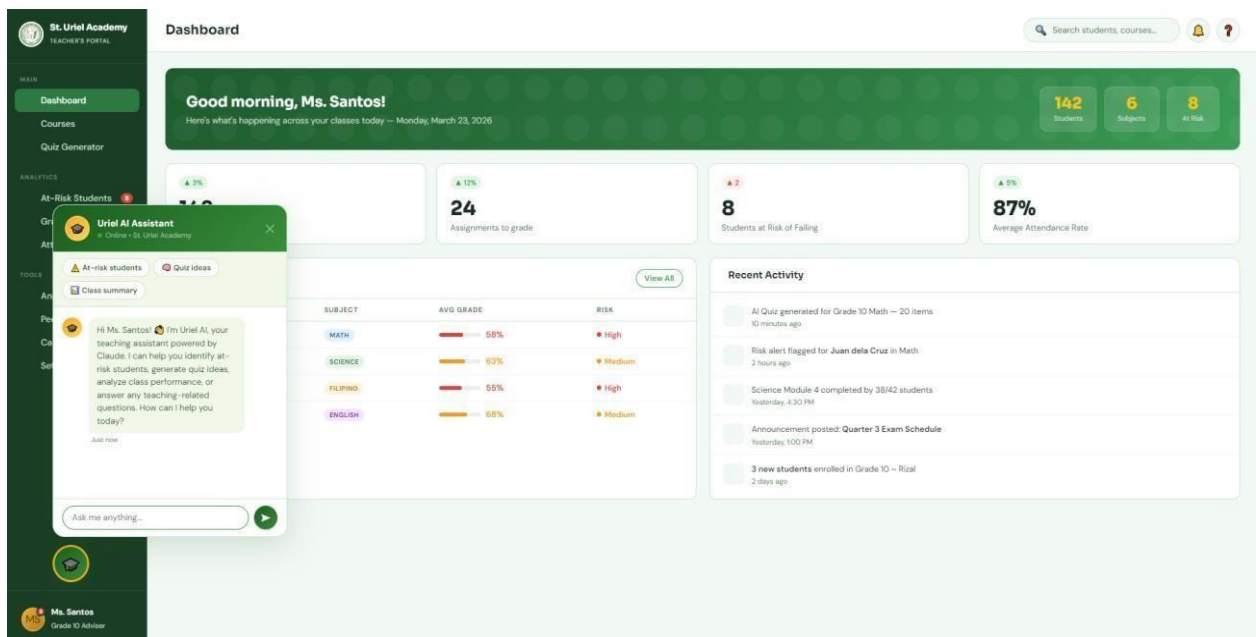


Figure 5.1:3 Teacher's AI Chat Head

The AI Chat Head provides real-time, context-aware academic support for teachers. The module allows users to ask questions related to teaching strategies, lesson planning, quiz creation, and explanations based on uploaded learning materials. It is restricted strictly to school-related content to ensure academic relevance and safety. This module enhances teaching efficiency by acting as an on-demand intelligent assistant that supports instructional tasks and decision-making.

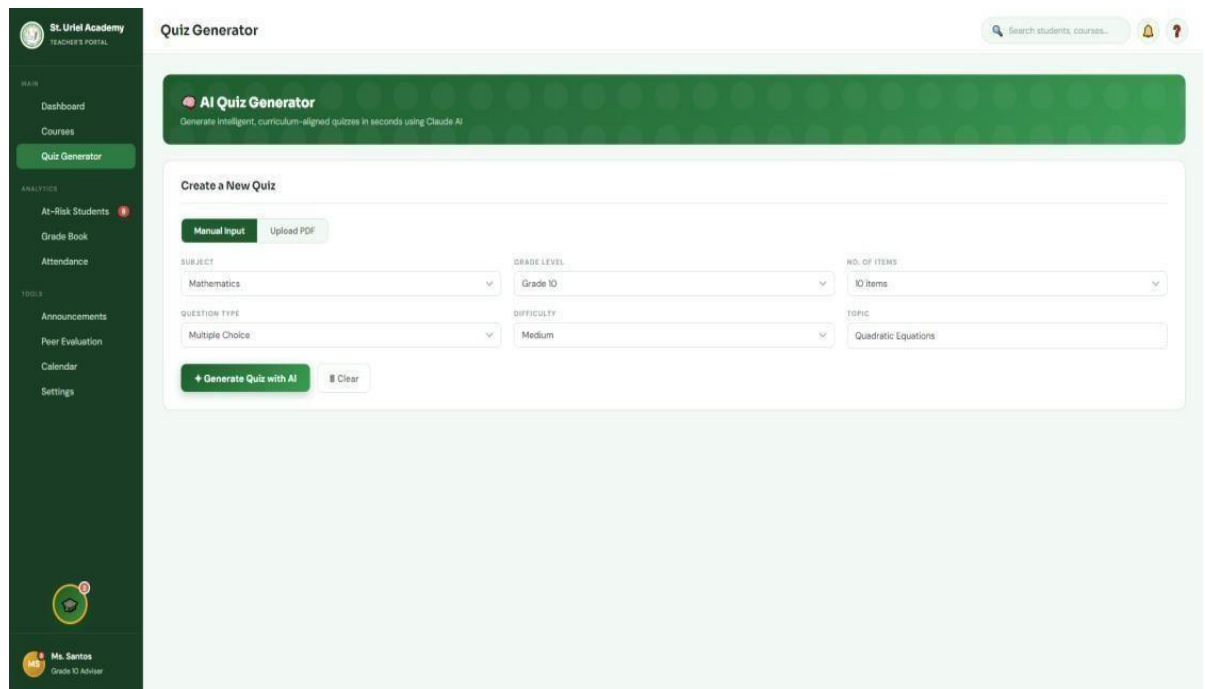


Figure 5.1.4: Teacher's Quiz Generator

The AI-Powered Quiz Generator module automatically creates quiz questions based on uploaded learning materials such as PDFs, presentations, and lecture notes. It helps teachers quickly generate relevant assessments aligned with lesson content and learning objectives. Teachers can adjust settings like difficulty level, number of questions, and time limits, and they may also edit the generated items. This module improves efficiency by reducing the time needed to create quizzes while ensuring relevant and consistent assessments.

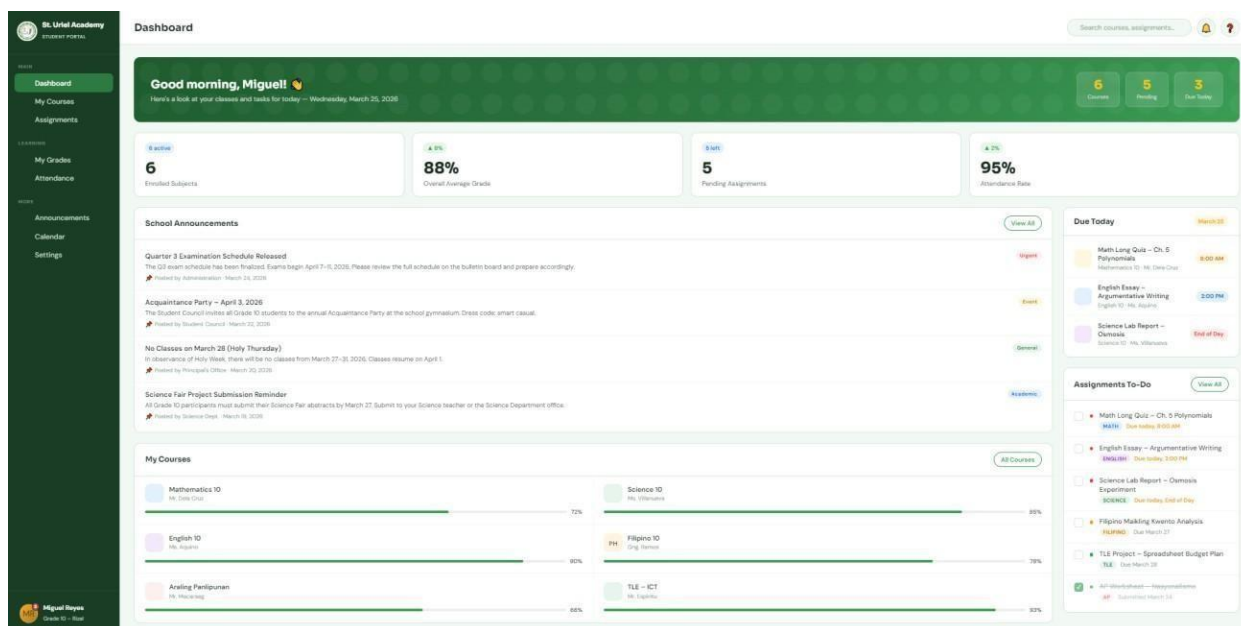


Figure 5.2: Student Dashboard

The Student Dashboard gives students a personalized, organized view of their academic life. The top section greets the student by name and shows summary cards: number of enrolled subjects, overall average grade, pending assignments, and attendance rate.

The main area is divided into three sections. The School Announcements section displays categorized posts (Urgent, Event, General, Academic) from teachers and administrators, replacing the need for Messenger group chats. The Due Today panel on the right highlights assignments and quizzes with their deadlines and subject color-coding, helping students prioritize their tasks. The My Courses section at the bottom shows each enrolled subject with the assigned teacher and a progress bar indicating course completion.

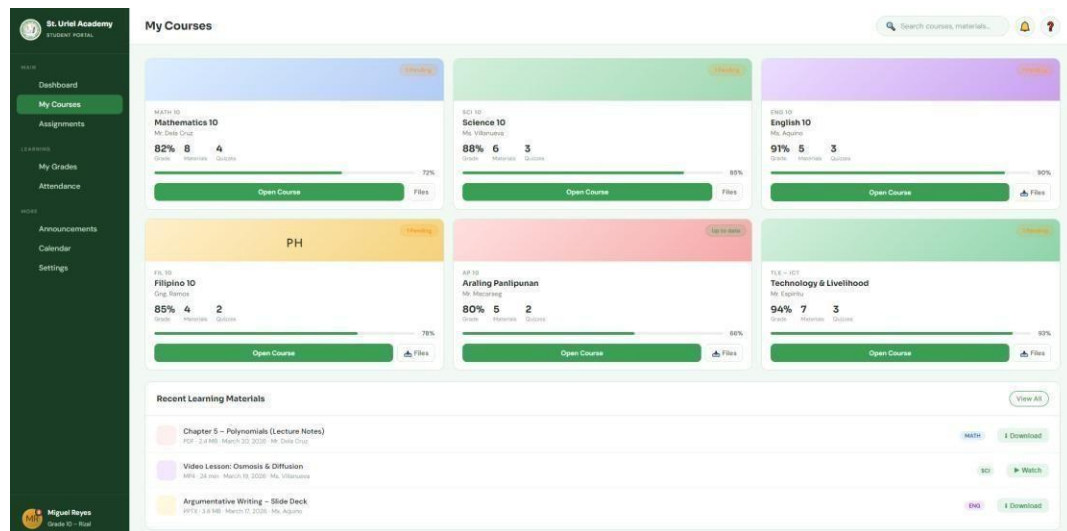


Figure 5.2.1: Student My Courses

The Student Courses module allows students of St Uriel academy of Taguig city to quickly view their enrolled subjects monitor their own progress, view recent learning materials uploads, and access course materials.

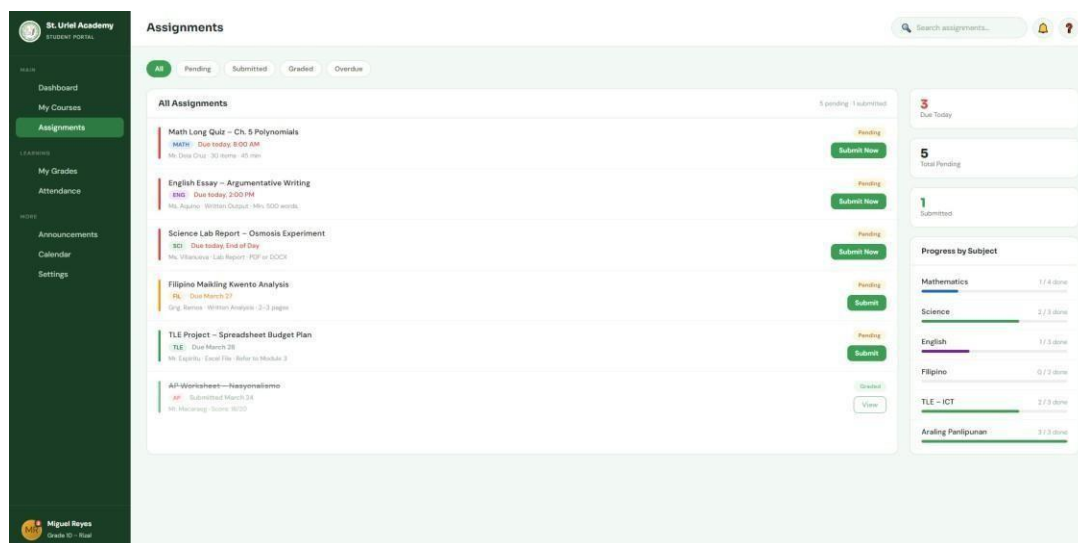


Figure 5.2.2: Student Assignments

The Assignments module provides students with a centralized view of all tasks assigned to them. It displays assignments by subject, along with key details such as titles, instructions, deadlines, and submission status.

The module helps students stay organized by highlighting pending and overdue tasks, often using visual indicators or color-coding for easy prioritization. It may also include filters or categories, allowing students to quickly sort out assignments based on subject or due date. Overall, the Assignments module supports time management and accountability by ensuring students can easily track, complete, and submit their work on time.

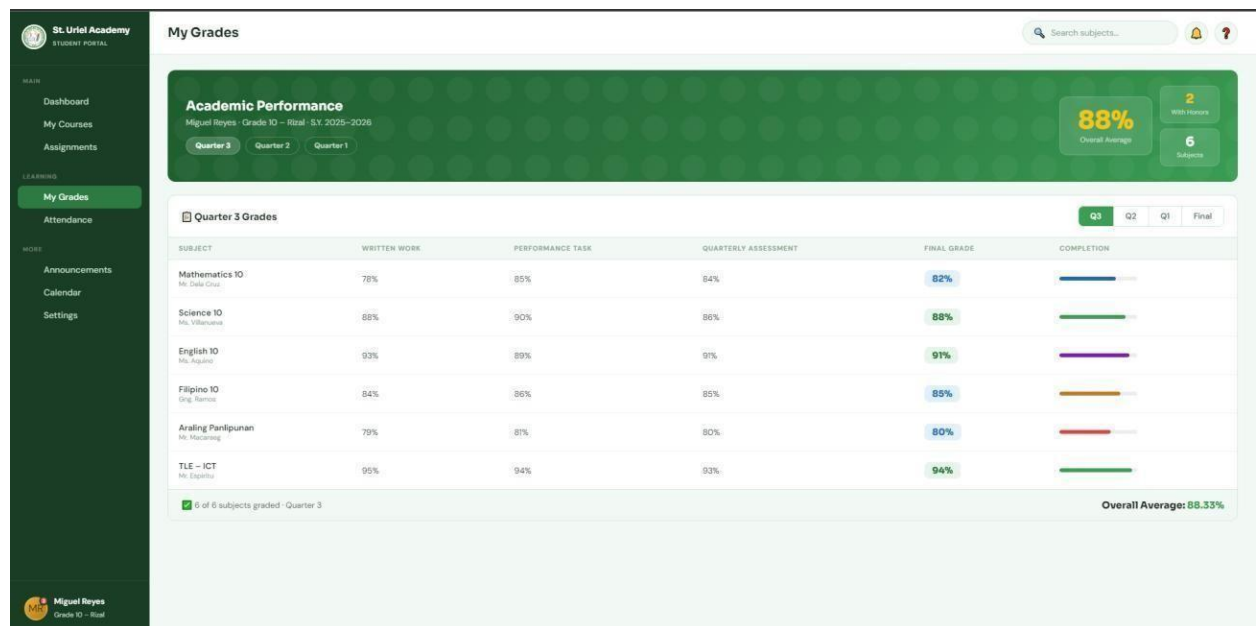


Figure 5.2.3: Student My Grades

The My Grades module gives students a clear and organized view of their academic performance. It displays the overall average, number of subjects, and honors status at the top for quick monitoring.

The main section shows a detailed breakdown of grades per subject, including written work, performance tasks, quarterly assessments, and final grades. Color-coded indicators help students easily identify their strengths and areas for improvement.

Students can also switch between quarters to track their progress over time, making the module a useful tool for monitoring and improving academic performance.

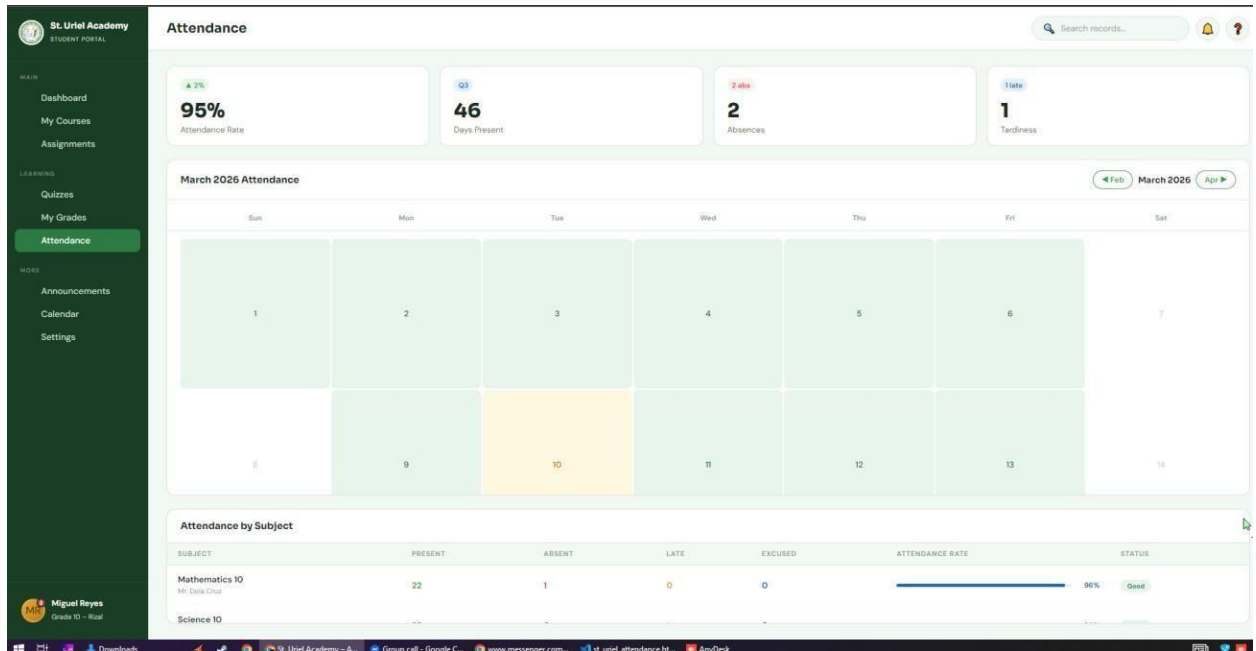


Figure 5.2.4: Student Attendance

The Student Attendance module allows the tracking and monitoring of students' daily attendance in an organized and efficient way. It records whether a student is present, absent, or late, and provides a clear overview of their attendance status.

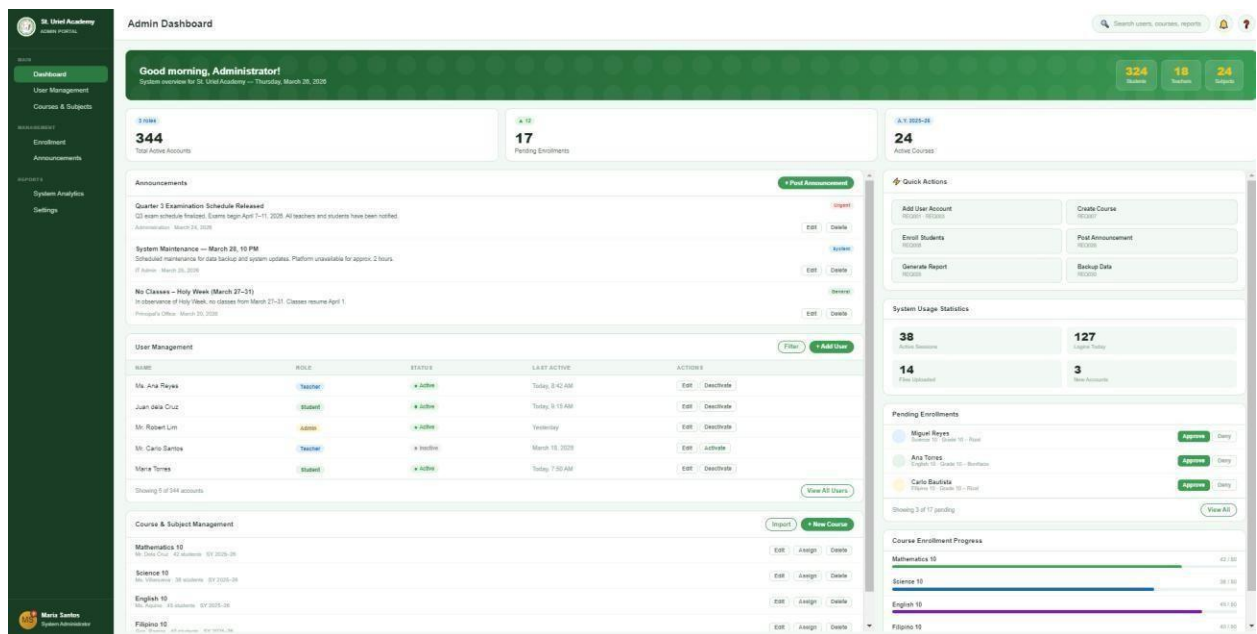


Figure 5.3: Admin Dashboard

The Admin Dashboard gives the system administrator comprehensive details of the entire institution's activities and information. At the top there are three institution-wide counts: total students, teachers, and active courses. There are four main panels inside the admin dashboard: the Announcement panel for uploading, updating, and deleting announcements; the User Management panel, which will assist the admin to quickly add, deactivate, and update user accounts; the Course & Subject panel for managing courses and subjects; and Quick Actions, which allow the admin to perform quick actions such as adding users and courses, enrolling students, uploading announcements, generating reports, and backing up data.

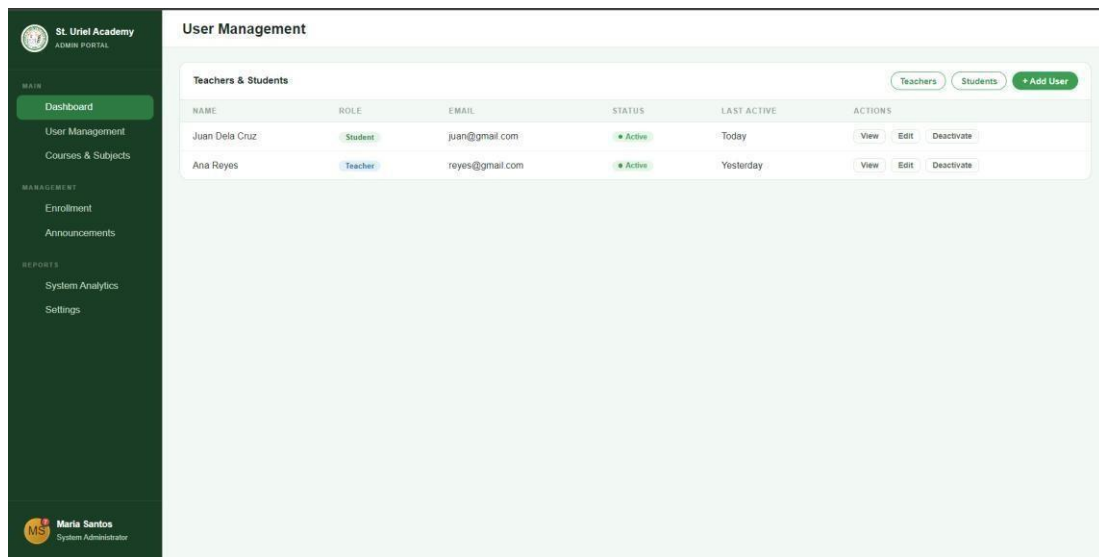


Figure 5.3.1: Admin User Management

In the User Management module, administrators are granted in managing system users by adding, editing, or removing accounts. It also lets them assign roles, control access, and monitor user status. This module ensures organized and secure handling of all users within the system.

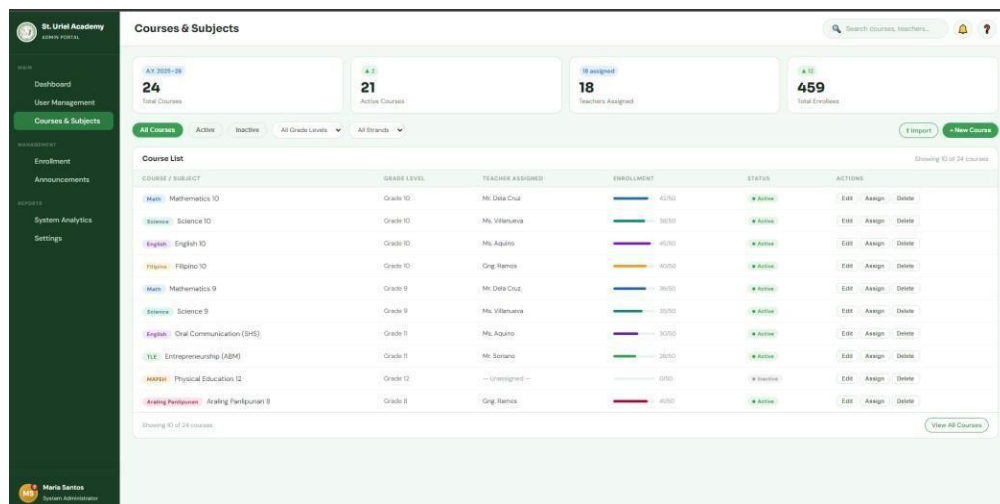


Figure 5.3.2: Admin Courses and Subjects

The Courses and Subjects Management module allows administrators to create, update, and manage courses, assign teachers, and handle student enrollment. It supports role-based access to ensure that only authorized users can modify course data, while teachers and students interact with assigned subjects.

St. Uriel Academy
ADMIN PORTAL

MAIN

Dashboard

User Management

Courses & Subjects

MANAGEMENT

Enrollment

Announcements

REPORTS

System Analytics

Settings

100%

Maria Santos

System Administrator

Enrollment

12

Pending Requests

38

Total Enrolled

This week

42

Approved This Week

3

Denied / Withdrawn

Pending (17)

Approved

Denied

All Records

All Grade Levels

All Courses

Enroll Student

Pending Enrollment Requests

Approve All

	STUDENT NAME	GRADE LEVEL	COURSE REQUESTED	DATE SUBMITTED	STATUS	ACTIONS
☐	Miguel Reyes	Grade 10	Science 10	March 25, 2025	Pending	Approve Deny
☐	Ana Torres	Grade 10	English 10	March 25, 2025	Pending	Approve Deny
☐	Carlo Bautista	Grade 10	Filipino 10	March 25, 2025	Pending	Approve Deny
☐	Liza Mendoza	Grade 9	Science 9	March 25, 2025	Pending	Approve Deny
☐	Jose Fernandez	Grade 11	Entrepreneurship (ABM)	March 27, 2025	Pending	Approve Deny

Showing 5 pending requests

Export CSV

Figure 5.3.3: Admin Enrollment

The Admin Enrollment module manages student enrollment requests, allowing administrators to review, approve, or deny applications. It provides filtering options such as grade level and course thus streamlines the process of assigning students to subjects, ensuring efficient and organized enrollment management.

REFERENCES

- Abaricia, C., & Delos Santos, M. L. (2023). Enhancing e-learning system through learning management system (LMS) technologies: Reshape the learner experience. *arXiv*.
<https://arxiv.org/abs/2309.12354>
- Alenezi, M. (2023). Digital learning and digital institution in higher education. *Education Sciences*, 13(1), 88. <https://doi.org/10.3390/educsci13010088>
- Asri Siti Fatimah, Bilqis Nurfitriani, & Ruslan. (2022). Students' perception of the use of Canvas-based learning activities in online classroom: Focusing on its benefits. *ResearchGate*. <https://www.researchgate.net/publication/362464613>
- Cardona, C. R., Ramirez V, G. M., Diaz, J., & Moreira, F. (2024). AI chatbot for teachers for an LMS platform. *Lecture Notes in Networks and Systems*, 307–317.
https://doi.org/10.1007/978-3-031-45645-9_29
- Embarak, O. H., & Hawarna, S. (2024). Automated AI-driven system for early detection of at-risk students. *Procedia Computer Science*, 231, 151–160.
<https://doi.org/10.1016/j.procs.2023.12.187>
- ICT-enabled information management system (IMS) in educational organizations. (2025, November 27). *IEEE Conference Publication | IEEE Xplore*.
<https://ieeexplore.ieee.org/abstract/document/11318254>
- Instructure Community. (n.d.). What is Canvas?
<https://community.instructure.com/en/kb/articles/662716-what-is-canvas>

- Jaca, C. A. L. (2022). Students' online class experiences using Google Classroom amidst the Covid-19 pandemic. *Randwick International of Education and Linguistics ScienceJournal*, 3(1), 58–68. <https://doi.org/10.47175/rielsj.v3i1.409>
- Levy, R., Asman, O., & Barnoy, S. (2024). Boundary-blurred behaviors in academic teachers-students facebook interaction: Are guidelines needed? A cross-sectional study. *BMC Nursing*, 23(1). <https://doi.org/10.1186/s12912-024-02466-y>
- Lopez, A., & Rufino Lorejo. (2023). Effectiveness of enhanced basic education information system (EBEIS)/ learner information system (LIS) data and school management. *Philpapers.org*. <https://philpapers.org/rec/LOPEOE-2>
- Mariappan, K., Singh Kaur, A. G., & Singh Kaur, P. B. (2024). The effectiveness of learning management system (LMS) as a digital tool for TESL undergraduates at higher education institution in Malaysia. *Journal of Social Sciences and Business*. <https://qiu.edu.my/journal/index.php/jssb/article/view/38sd>
- MoodleDocs. (2025, August 21). About Moodle. Moodle Documentation. https://docs.moodle.org/501/en/About_Moodle
- Muhisn, Z. A. A., Almansouri, S., Muhisn, S., Ahmad, M., & Omar, M. (2022). The effectiveness of knowledge combination in e-learning management system (eLMS). *International Journal of Emerging Technologies in Learning (IJET)*, 17(16), 33–42. <https://doi.org/10.3991/ijet.v17i16.31873>

- Samsyudin, I. (2025). Native apps vs. progressive web apps: A comparative analysis of user experience and development costs. *SSRN*.
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5039280
- Veseli, A., Hasanaj, P., & Bajraktari, A. (2025). Perceptions of organizational change readiness for sustainable digital transformation: Insights from learning management system projects in higher education institutions. *Sustainability*, 17(2), 619.
<https://doi.org/10.3390/su17020619>
- Vijoriya, A. S., Parmar, Y., & Jatav, B. S. (2026). Comprehensive analysis of ICT-based learning management systems: Best practices for enhancing digital learning experiences. *In Lecture Notes in Networks and Systems* (pp. 355–364). https://doi.org/10.1007/978-3-032-06688-6_35
- Yidana, P., & Akuna, E. (2025). Academic records management practices in Africa higher education institutions: Empirical review. *British Journal of Contemporary Education*, 5(1), 93–113. https://abjournals.org/bjce/wp-content/uploads/sites/20/journal/published_paper/volume-5/issue-1/BJCE_JRYJWENT.pdf

APPENDICES

APPENDIX A: Resource Persons

Interview Transcript

Interview Date: 02/09/2026

Person to be Interviewed: Poul Luis C. Sarmiento

Position: Principal

Interview Transcript

02 February 2026 — St. Uriel Academy of Taguig City

Name	Transcript
Interviewer Jana	Good morning, good afternoon po.
Interviewer Jana	Yung pinaka gist po o yung idea po ng e-learning o yung ELMS is based po sa experience po namin. Una is kami pong student ng STI is dun po yung inspiration niya.
Interviewer Jana	So ang pinaka main problem po na illessen po natin dito is yung manual process on the perspective ng students, teachers, as well as you as a principal po of this school.
Interviewer Jana	So ang ELMS o e-learning management system, iisa-isain ko na lang po muna yung depende sa perspective. Sa student po, ang pwede po makita ng student is idedigitalize na lahat.
Interviewer Jana	Magiging transparent siya. First po is sa attendance. The student is able to know the attendance kung ilan.
Interviewer Jana	Pero depende po yun sa teachers kung ipapakita po nila dun sa student nila yung attendance ng each student. Sa student naman po, next is yung sa announcement. Ang main goal po dito is to separate the personal life of a student as well as the academic life.
Interviewer Jana	Kapag personal life po, of course, sinabi natin ang means of communication of the blended learning is mostly focusing on the messenger platform. Which is mahirap po siya kasi, syempre kami po as student, nahirapan na rin po kami sa mga group chat kahit na sa messenger. What more pa po if nandun yung mga resources of learning? Such as yung PPT, yung visuals, yung notes ng teachers, yung announcement.
Interviewer Jana	So hindi po siya masyadong napapansin, pero natatapan lang. Kaya nag-develop po tayo ng electronic learning management system ng school na St. Uriel is focusing po sa productivity ng students. Kumbaga, kapag sinabi po natin nag-lock in sila doon sa pinaka system natin, automatically productivity po ng school siya.
Interviewer Jana	Kasi ang makikita nyo lang po talaga doon is announcement, gawain, and mag-aral. So next naman po is yung sa transparency ng student, sa attendance, sa grades. In terms of activities, maaari pong ipakita ng mga teachers yung score mismo ng student para aware sila sa kung papaano po yung gagawin nila, kung kailangan na po ba nilang maghabol, kailangan na nilang mas galingan pa next activities.
Interviewer Jana	And since blended learning nga po tayo, may mga activities po na kahit naman po sa STI is applicable dito. For instance, may activity po na gusto pa rin ipagawa si teacher na on-site o face-

Name	Transcript
	to-face. Pwede pong ang gawin ni teacher is pen and paper yung quiz. So ang gagawin nalang po ni teacher, mag-open po siya ng dropbox doon sa ating system, and then after po yung open, is i-upload po ni student kasama po yung score and yung name niya. Para si teacher, kahit checkan po nila ng face-to-face or palitan lang po, makikita pa rin po ni teacher yung record ng student.
Interviewer Jana	But, dako na po tayo doon sa teacher's perspective. Kapag si teacher's perspective, first and foremost po na binibigyan niya is convenience. Kapag convenience po kasi, of course, sobrang hirap po mag-check ng manual process. Hindi naman po siya typical na sobrang hirap kasi ganito, pero mas makapadali po siya ng digitalization. Kasi kapag manual process po, yung papers at some point pwedeng mawala, pwedeng makalimutan, and wala pong proof si teacher na gano'n talaga.
Interviewer Jana	Sabihin natin yung sa deadlines po ng students. For instance, pwede pong, quote-unquote, i-gaslight ng student na pinasa ko po yung answer on time. But with the help of our system, si teacher may choice kung i-allow niya yung late submission doon, pero kitang-kita niya real-time po yung pagpasa kahit na reset niya yung deadline, nakikita niya pa rin na late, which is mag-grade na niya ng maayos.
Interviewer Jana	Specifically, kapag mayroon po sa kriteria ng late submission is deduction. Next naman po sa teacher's perspective is checking of attendance. Hindi na po kailangan magsulat ng mga pangalan sa mga yellow pad, or sabihin natin intermediate pad to know the attendance of the student. But with the help of our system, mas madi-digitalize and centralize po.
Interviewer Jana	As dako naman po tayo sa perspective as principal of this school, you can handle po directly yung mga teacher's evaluation with the help of our system. For instance, you want to know who are the teachers that are performing well by having the teacher's evaluation po. Nasasagutan din po ng students. Like kung sino po yung mga doing good na teachers or sa mga feedbacks ng ganoon.
Interviewer Jana	But sa principal po, para hindi po masyadong nakaka-bombard yung gagawin, teachers lang po yung pwede niyong i-perspective. Pero pwede naman po directly sa student, pero yung sa mga announcement po, depende pa rin po sa teachers. And of course, yung IT po natin, sa kanila po mag-focus yung, sabihin natin help desk, yung mga accounts po. So, kung narin, sir, hindi ko po mag-login yung account po. Paano po yung ganoon? Si IT na po yung mag-handle.
Interviewer Jana	So, yun lang po yung brief discussion about the system.
Principal	So, meaning, each student will have their own account. And who will do that? Kayo?
Interviewer Jana	Opo.
Principal	Tapos, yung sa ano, hindi ko nalang sabihing na, let's say for example, nagkaroon ng activities sa LMS. Tapos, di ba automatic yung score yun?
Interviewer Jana	Apo.
Principal	Ano yung recording na rin yun sa, parang, may class record ba yun sa loob mismo yung teacher?
Interviewer Paller	Bali, sir, yung class record po pa rin, pero malalaman niyo po kung saan yung, kung kunin niyo yung grades nito, yung parang, nariyo pa yung paper na ganito para record yun. So, check-in na lang po yung LMS para i-record na lang po yung mga grades po na nakala po ng system po, sir.
Principal	Meaning, i-input pa rin?
Interviewer Jana	Sa Excel, apo. For instance po, sa platform ng Excel, ganoon.
Principal	Pero yung LMS niyo, pwede doon yung natouch na activity.
Interviewer Jana	Oh, okay. Merong parang... Dropbox po, parang...

Name	Transcript
Principal	Hindi, yung ano, for example, di ba yung may questions, tapos may bitunditin lang, tapos next number, tapos makita mo yung score.
Interviewer Jana	Yes, po.
Principal	Yung score, di siya automatic na re-record doon?
Interviewer Paller	It depends po, sir, kung pre-defined na po. Kaya yung quiz lang naman po is about true or false, kung may mga specific na sagot. Automatic, kita na po ng system yun, sir. Pero kapag mga essay po, sir, or mga kailangang evaluate na naman yung teacher...
Interviewer Jana	Si teacher po yung magbibigay ng score.
Principal	Pero, may automatic recorded yun.
Interviewer Paller	Opo, sir.
Principal	Doon sa system na... Parang pinaka-classify po yung teacher, makikita.
Interviewer Paller	Apo. Sir, bali mag-check po tayo sa mga questions ka po ngayon, sir.
Interviewer Alayan	Sa school background and population po tayo, kailan po naitatag ang Sint or Nakalimutali City at anong mga grade levels ang in-offer ng school?
Principal	So, June 2005. But I do not know the exact date, but June 2005. And then, kindergarten to... pre-school to senior high school.
Interviewer Alayan	So, ilan naman po yung kabuuan ng bilang ng students sa junior high school and senior high school po? Sa junior high school.
Principal	Sa junior high school, we have 199.
Principal	And senior high school is 216.
Interviewer Alayan	Next question po. Ilan po ang sections sa junior high school at ano-anong pong strand ang in-offer sa senior high school?
Principal	So, in junior high school, we have 107. There are six classes. And then, sa senior high school, we also have six classes.
Interviewer Alayan	Ano-ano pong strand ang in-offer?
Principal	ABM, GAS, HUMMS, and techVoc ICT.
Interviewer Alayan	Kamusta po ang student population ng school sa mga nakaraang taon? Ito po ba ay tumaas, bumaba, o nanatiling stable?
Principal	Tumaas. Ngayon, nasa 613.
Interviewer Alayan	Kamusta naman po ang graduation success rate ng school sa junior high school at senior high?
Principal	No 100 percent. Kasi there are school districts. Especially those who decided to stop. Mga 1 percent, 2 percent. Mga 98 or 99 percent.
Interviewer Alayan	Next po, about naman po sa teachers and academic structure. Ilan po ang teachers na mahawak ng junior high school at senior high school?
Principal	So junior high school, we have 6. Then senior high is 10. So total of 6, 10 teachers.

Name	Transcript
Interviewer Alayan	Paano po ang grading system ng junior high school at senior high school?
Principal	Mayroon kami sa way din ano. Computation. We did not follow the depEd computation. And computation na yun, is a habit na kami.
Interviewer Paller	May bawal idisclose po yun.
Principal	Oo. Bawal lang yun.
Interviewer Alayan	Next po. Sa pag-a-assign po ng teachers sa sections, subjects at schedules, paano po ito minamanager at saan po ini-store ang records?
Principal	Class program.
Principal	Kaya making a class program of each teacher. Yes, po.
Principal	Ako kasi gumagawa, nag-drop lang ako. Table, time, mga subjects, after pang na, let's say for example, for grade 1, 7 to 11:30, to 9. So, ilalagay ko yung mga subjects. Pag nalagay ko ng subjects, tapos na time naman ilalagay ko yung kung sino yung mga teacher. Na hahandle na teacher. Grade 2, ganoon din.
Interviewer Paller	Bale, Sir, gumagamit po ba kayo ng software doon, like Excel or paper po? Tapos, yung paper tapos, yung inkod sa Excel.
Interviewer Alayan	Next, about ako po sa Existing Systems and Data Management. May mga existing system po ba na ginagamit ang school para sa mga proseso o transactions?
Principal	Manual. Manual.
Interviewer Alayan	Ano-ano pong software o applications ang ginagamit ng St. Uriel sa pagmamanage ng student records?
Principal	We do not have the system kasi, it's like the SF thing. We just print up. We just print the form and we write the data. Then, ilalagay sa isang folder at ito tapos sa cabinet. So, manual pa po.
Interviewer Alayan	Ang students dito po ba ay digital stored or naka paper records ka po?
Principal	Paper. Pero, kung sa paper records na siya, hard copy. Hard copy. Pero, sa same time, mayroon na yung soft copy.
Interviewer Alayan	Kailan pong nag-start na magkaroon ng soft copy?
Principal	2019. Kasi, ginagamit yung form yung ina-download namin. Kasi, mayroon SF 10 na 980. Tapos, isi-save lang namin. Tapos, i-print up. Kasi, ilalagay ng print.
Interviewer Alayan	Will the school allow the system to analyze historical student data for academic purposes?
Principal	School will allow na?
Interviewer Jana/Alayan	Historical student data. Ang bagay yung mga past po ng students para malaman kung sino po yung at-risk at failing. Yung mga hawak nyo na po ng records ngayon is i-input po natin doon sa systems.
Principal	Kahit wala na dito yung mga. Ay, hindi. Oo naman.

Name	Transcript
Interviewer Alayan	Next po. About teaching, monitoring, and school management. Bilang principal po, paano po ninyo dinidistribute ang mga modules at announcements sa teachers?
Principal	We do a lot of modules. Kasi, mayroon mayroon kami yung books.
Interviewer Paller	Yung sa senior high po, sir? Sa senior high. No modules. The teachers will provide yan. The activities could be kasi mayroon ginagamit printed, usually printed, or unprojected. Yung activities.
Principal	Hindi. Mayroon mayroon powerpoint ng teacher. So, it depends sa... Powerpoints would be given to the students. At least, they will not get referenced by the teachers.
Interviewer Jana	By having group chats po, ganun?
Principal	Oo.
Principal	Syempre, they need to buy. Each time, bibili sa TLC.
Principal	Hindi, lima lang. Tapos, ang the rest, bibilhin na nila.
Principal	Syempre, 75 percent.
Interviewer Alayan	Next po, paano po ninyo minimonitor ang performance ng teachers pagdating sa kanyang lang pag tuturo?
Principal	I have this one. I ask random students. Para yun, informal na mayroon kasi, halimbawa, observe. Tapos, tatawag. Tapos, sa mga teacher, sinong teacher mo, may ginawa, anong ginagawa ng teacher mo ganon. Pero, I also do class observation before end of the class. Nagkakasala ng class demonstration.
Interviewer Alayan	Sa ngayon po ba, may hinahawakan po ba kayong section na tinuturuan?
Principal	Yes grade 12.
Interviewer Alayan	Next, tungkol naman po sa learning materials and attendance and submissions, paano po dinidistribute ang learning materials at announcements sa students po?
Principal	By announcing through gc messenger.
Interviewer Alayan	Next po paano po nagsusubmit yung mga students sa kanila pong mga assignments?
Principal	In person kasi wala naman kami ELMS.
Interviewer Alayan	Sa attendance naman, paano ito nire-record?
Principal	Ganon din using SF forms. It's a form for depEd.
Interviewer Alayan	May mga difficulties ba kayo na-encounter sa pag-manage ng learning materials?
Principal	Kasi books can considered as learning material if the students naman has books kasi kung walang reference ayun doon magkaroon ng problema we need to prove for students that we have reference.
Interviewer Alayan	Ano ang mga common challenges sa student submissions at late?

Name	Transcript
Principal	Lates no quality for clients aminin nyo.
Interviewer Alayan	May pagkakataon ba na nalilimutan ng students ang announcements or deadlines?
Principal	They are some those who are not willing to submit.
Interviewer Alayan	Next tungkol naman po sa workload and automation needs ano po yung pinaka time consuming na ginagawa nyo as principal na sana ay ma-automate.
Principal	Sa grades kasi namin meron kami form soft copy excel ayon kada input automatic na sya okay naman sya.
Interviewer Paller	Ano sir sa pagkolekta po ng mga activity or pag score po ng mga essay po ng mga students like assignments po nila for example tapos napo yung period nato may mga time po ba na nawawala yung paper.
Principal	Kasi collect and after collect check and return. Hindi kami nag sstore sa mga computer.
Interviewer Alayan	Gaano kayo kakomportabel sa paggamit ng computer-based o online system?
Principal	Syempre if I don't have idea about your system it will be difficult for me pero if I'm using the system syempre diba we can adjust encode encode lang ppt.
Interviewer Alayan	Next po sa AI integration and ELMS perception ano po ang perspective ninyo sa pag integrate ng AI bilang assistance ng teachers.
Principal	Nakakatulong sya of course, kahit ako gumagamit nyan e, so if merong ganyan yung system na gagawin nyo much better, d ako against dyan.
Principal	Actually ano ako, kinoconsider ko ung AI sa messenger yung AI ba yun. Specially where looking for activities. Kaso ngalang ang downside non you need to check the questions if its relevant na hindi naturo alam nyi naman yung mga studyante nagrereklamo kapag hindi naturo, like sir bakit may ganito wala naman to sa tinuro nyo yun lang naman when it comes for creating a question mabilis.
Interviewer Jana	Ano po sir some features po ng AI na i-integrate po namin is yung pag analyze po kung sino po ung mga student po na risk of failing instance of computation itself sinabi nyo po kanina na nag encode po kayo sa excel is automatically nya pong nacompute yung grade sa AI naman po pag sinend po natin yung excel file po don sa model or AI natin mapipili nya napo kung sino po yung may lower than 75 na grades po.
Principal	Parang E-class record i-upload don sa system tapos yung system na ang bahala.
Interviewer Jana	Opo dun napo machecheck kung sino po yung mga student na babagsak.
Principal	Ano yun may load ago, halimbawa yung file ng excel upload mo lng yon?
Interviewer Paller	Opo sir ma-analyze napo nya yun.
Principal	Talaga?
Interviewer Jana	And of course po sa analytics din po is yung performance po ng students na 75 percent below yung grade nya.
Principal	Ano bayung ano don makikita is theres a remark ba?
Interviewer Paller	Yes po sir for example po napansin po ng AI na medyo mababa po yung mga seatwork nya

Name	Transcript
	sasabihin po ng AI na kausapin napo yung studyante nato kung bakit po ganon yung grades nya po.
Principal	Teachers lng ba makakita ng AI?
Interviewer Paller	Yes po.
Principal	Ano yun parang notification?
Interviewer Paller	Ano lang po yun sir para lang po syang bubble yung chat head po sa messenger parang ganon po sir tapos iclick nya po sya sir tapos mag pop up po ung AI na yun tapos pwede napo mag upload don.
Principal	Ahh.
Interviewer Alayan	Sa tingin ninyo po ba nakakatulong po ba yung ELMS with AI na kayang mag summarize ng lesson plans, mag generate ng quiz suggestions, magbigay ng teaching recommendation at makakita ng students at failing.
Principal	In terms sa lesson plan pano yun?
Interviewer Jana	The AI will summarize the lesson plan for the teacher, for instance the AI will give suggestions kung ano po yung mga activities na pwedeng ipaga ng teachers na align po don sa lesson plan na ibibigay parin po ni teacher. Like for instance po ang itatale is about sabihin natin ICT program magbibigay po ng paonting exercise na ibibigay ni AI when it terms of programming po sa mga students.
Principal	Ahh they can only get activities walang problema don mas paborito pa nga ng teacher yun ung quiz and teaching recommendations and lesson.
Interviewer Alayan	Next po naniniwala po ba kayo na ang ELMS with AI assistance for teacher support ay makakatulong sa pagpapabuti ng student and balance?
Principal	If tama yung pag gamit and viniverify ng teacher yung generated output bago i distribute sa students, then yes. If basta nalang mag generate without verifying, dun na yung panget na part.
Principal	Somehow kasi for example the teacher uses the recommended activity meaning to say the performance of the student, pero not 100 percent. Tanong ko lang ELMS with AI naba?
Interviewer Paller/Jana	Opo si lr sa teacher side po sa student po bawal mainly activities lang po talaga pero sobrang helpful nya po in a way po na hindi napo namin kaialangan mag notes kasi announcement po and quizzes are taken online po. And hindi narin po kaialangan ng backread sa messenger po kung nasaan po ung mga resources like handout. Sa ELMS po nakalistdown napo lahat lahat.
Principal	Saan galing yung handouts?
Interviewer Paller	The teacher po pwede po sila mag upload ng mga materials po nila.
Principal	Yes.
Interviewer Alayan	Papayagan nyo po ba kami gumamit ng school data para po sa academic development of system?
Principal	Student academic like?
Interviewer Paller	Like for creating an account and grades po nila ganon po sir for the academic po we will ensure secure po yung privacy, data po nila sa amin.
Principal	Yes, pero yung numbers of students and teachers and depende sa mapag decidedan ko na pwede

Name	Transcript
	nyong gamitin.
Interviewer	Handa po ba ang school na i-adopt ang proposed system?
Principal	Oo naman.
Interviewer	Mayroon po ba kayong expectations na dapat naming isaalang-alang sa pagdevelop ng system?
Principal	Uhm, ang pinaka expectation ko lang is confidentiality, mas better if yung data na ma store sa system is only accessible to limited people lang and also walang risk na ma leak dahil sa system malfunction.
Interviewer	Inaaprubahan nyo po ba yung development ng SUA: IntelliLearn sa St Uriel Academy of Taguig City?
Principal	Yes

Interview Transcript

Interview Date: 02/07/2026

Person to be Interviewed: Noel A. Baga Jr.

Position: Teacher/IT Admin

Name	Transcript
Interviewer Alayan	Bali sir para po sa first question po namin, may mga existing System po ba na ginagamit ang Uriel regarding po sa mga processes or transactions ng school po?
Employee	Ahh wala, bali manual lang talaga mostly.
Interviewer Alayan	Sa ngayong proseso po paano nyo po minamanage ung mga lesson materials nyo po? Like ung mga handouts, modules, activities po? Like san nyo po iniistore?
Employee	So, sa pagdistribute ng handout at activities messenger, natatabunan nga minsan haha.
Interviewer Alayan	Paano nyo naman po binibigay ung mga modules, activities, quizzes, and assignments sa mga students nyo po?
Employee	Ganon lang din tapos sa mga quizzes at mga activities gagawin nila sa phone at screenshot at ipapasa sa messenger.
Interviewer Alayan	Paano nyo po binibigay or disseminate announcements regarding sa mga announcements nyo po?
Employee	Sa messenger lang din.
Interviewer Alayan	Paano nyo naman po na momonitor and nirerecord ung performance ng mga students nyo po?
Employee	Gumagamit kami ng excel sa halos lahat ng performance monitoring, grades, scores at kung ano-ano pa.
Interviewer Alayan	Ano naman po ung mga software or tools na ginagamit nyo for this kind of process?
Employee	Ayon, Excel.
Interviewer Alayan	Ano ano po ba ung mga challenges na napapagdaanan nyo po regarding sa mga students' submissions and grades?
Employee	Minsan by paper or messenger natatabunan yung chat tapos minsan nawawala yung papel kaya yung mga student minsan naamarkahan ng late or no submission kasi hindi na makita yung output nila kasi nalilimutan kung saan nakalagay yung mga papel at natatabunan yung chat sa messenger.
Interviewer Alayan	May mga pangyayari po ba sir na nalilimutan ng mga students nyo po ung mga announcements nyo po or mga deadlines?
Employee	Oo, lagi kasi salin dila lang makakalimutan talaga yun or minsan nakapost sa bulletin board yung announcement pero napupunit lang din naman tapos minsan as usual ayun natatabunan lang din sa messenger.
Interviewer Alayan	Bukod po sa pagiging irresponsible ng students meron po ba kayong ideal factors na sa tingin nyo po nalilimutan ng mga students ung mga announcements and deadlines

Name	Transcript
	nyo po?
Employee	Yung ibang student nagta-trabaho tapos yung iba walang stable internet connection tapos minsan yung pinag-assign na magkalat ng announcement nakakalimot din, lalo na yung mga students.
Interviewer Alayan	Ano po ba ung most time consuming na task na ginagawa nyo as a teacher na gusto nyo po ma automate?
Employee	Yung pinaka-time-consuming talaga is yung pag-check, submission tapos pag-encode ng grades mano-mano kasi. Ayan gusto ko ma-automate ang mga yan.
Interviewer Alayan	Kayo po ba sir personally gumagamit po ba kayo ren po ng mga AI tools? Paano po eto nakakatulong sainyo?
Employee	Gumagamit ako syempre sa paggawa ng lessons pero hindi lahat ah. Assistant sa paggawa ng quiz at activities ayun na yon.
Interviewer Alayan	Sa tingin nyo po ba makakatulong po sainyo kapag merong ELMS na may integrated AI na kaya mag summarize ng mga lesson plans, generate ng mga quiz suggestions, provide teaching recommendations and spot students that at risk of failing?
Employee	Laking tulong talaga yan kasi halos full packaged na e. Kasi halos lahat ng kailangan ko nanjan na.
Interviewer Alayan	For the information ng lahat ng naka enrolled and ung mga past student's ng Uriel saan nyo po sya iniistore?
Employee	Old ways, store agad sa folder or cabinet.
Interviewer Alayan	Would the ICT department support the deployment of this system for academic use?
Employee	Oo naman sino ba may ayaw sa ganyan.

APPENDIX B: Letter of Permission (signed by principal)

APPENDIX C: Curriculum Vitae of the Client



Noel A. Baga Jr.

INFORMATION TECHNOLOGY

Contact Me

Blk 30 Lot 7 Tarhata St.
Pembo Makati City
1218

jhaybaga48@gmail.com

09362875730

Education

Bachelor of Science in Information Technology

ABE CUBAO COLLEGE
2015 - 2017

Bachelor of Science in
Information Technology
(STI) System technology Institute
Global City
2010 -2013

Web Design

Excel Technical Skills and Career
Center Inc.
2021

SKILLS

- Installing software or hardware
- Maintaining and repairing equipment
- Troubleshooting different computer issues
- Determining and installing appropriate security measures
- Basic in Programming: C, C++, Java, HTML, CSS

Work Experience

Travel Associate – Back Office October 22- Up to present
IGT Solutions | World Wide Corporate Center Shaw

DUTIES AND RESPONSIBILITIES

- Assists the appropriate department in providing a copy of the IOR.
- Helps identify trends of issues that need to be addressed in connection to security and safety.
- IOR Analyst monitor and respond to email inquiries and requests in the KCM mailbox.
- IOR Analyst verifies employee status to reflect as "Active" and qualifications and documents are current and updated.

PART TIME - IT ADMIN/INSTRUCTOR June 2017 - Up to present
St. Uriel Academy of Taguig City

DUTIES AND RESPONSIBILITIES

IT ADMIN

- Provide Technical Support
- Troubleshooting of computer hardware/software
- Provide Network support
- Online remote assistance

INSTRUCTOR

Mobile Computing – hardware and software
Webprogramming – HTML, CSS

Records Management Staff April 2019 - April 11 2020
Polyserve Inc/Phinma Properties
Mandaluyong Barangka Ilaya

DUTIES AND RESPONSIBILITIES

- Scanning of Client Documents
- Data Encoding of Client Information

REFERENCES

Ronnel Cagurangan

Supervisor - Data production division
GDPS
Phone:
Email : rcagurangan@geodata.com.ph

Leonel Allan J. Apor

Application developer| Accenture
Phone: 09294399560
Email : leonelapor88@gmail.com